IoT-Based Toxic Gas Detection and Monitoring System for Manholes and Workers

A Project Report

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ABSTRACT

This report outlines a comprehensive IoT-based system to ensure the safety of maintenance workers in manholes by continuously monitoring their health, the manhole environment, and the structural status of the manhole itself. The system consists of three primary components: health monitoring, environmental assessment, and manhole lid monitoring. Wearable sensors track workers' vital signs, such as heart rate, body temperature, and oxygen levels, providing real-time insights into their health. Simultaneously, gas sensors inside the manhole detect harmful gases typically methane. Additionally, a lid monitoring system tracks water levels within the manhole and detects any lid tilting, indicating potential flooding or structural instability. If abnormal health readings, toxic gas levels, high water levels, or lid movements are detected, the system sends immediate alerts to relevant authorities, facilitating rapid intervention. This integrated approach aims to significantly enhance the safety and response capabilities for manhole maintenance operations by addressing both worker well-being and environmental hazards.

CONTENTS

LISTOFFIGURES

LISTOFABBREVATIONS

IOT	Internet Of Things
ECG	Electrocardiography
PPG	Electrocardiography
GSM	Global System for Mobile Communications.
WiFi	Wireless Fidelity
MEMS	Micro Electro Mechanical Systems
SMS	Short Message Service
RAM	Random Access Memory
KB	KiloByte
HVAC	Heating, Ventilation, and Air Conditioning

CHAPTER 1

INTRODUCTION

Urban infrastructures rely heavily on manholes, sewers, and other confined spaces to house critical utilities such as water pipelines, electrical cables, and drainage systems. These spaces, while essential, pose serious health risks to workers who need to enter them for inspection, maintenance, or repair. One of the most dangerous threats is the presence of toxic and hazardous gases, including methane, hydrogen sulfide, carbon monoxide, and a lack of oxygen. These gases can accumulate without warning and often cannot be detected by human senses, leading to fatal accidents. Exposure to even small concentrations of these gases can cause dizziness, suffocation, or, in extreme cases, death.

Traditionally, the detection of such gases in manholes and other confined spaces has relied on manual monitoring techniques, including portable gas detectors carried by workers[1]. While these devices provide some level of protection, they are prone to human error, require constant vigilance, and do not offer continuous real-time monitoring[2]. If the worker is already inside the confined space when the gas levels become dangerous, the time available for evacuation is significantly reduced. This highlights the critical need for a more reliable and automated solution to monitor and detect toxic gases before workers are exposed. In recent years, the rapid advancement of the Internet of Things has enabled the development of smarter, automated systems that can address these challenges. IoT technology allows for the continuous collection, processing, and transmission of data from a network of connected devices[3][4]. In the case of confined space safety, an IoT-based gas detection system can provide real-time monitoring of gas levels inside manholes, offering immediate alerts when hazardous conditions are detected.

The system can be integrated with wireless communication technologies such as GSM or Wi-Fi, ensuring that data is transmitted to a cloud platform or mobile devices in real-time[5]. This allows for remote monitoring by supervisors and provides workers with immediate warnings through mobile or wearable devices, giving them enough time to evacuate.

The proposed IoT-based toxic gas detection system for manholes and workers aims to enhance safety by continuously monitoring multiple hazardous gas levels. By utilizing gas sensors installed within the manhole, the system collects data on the concentration of these gases and processes it through a micro controller. The data is then transmitted wirelessly to a centralized platform where it is logged, analyzed, and displayed in real-time. The system can also send alerts to both workers on-site and supervisors remotely through mobile applications or web-based dashboards, ensuring a proactive response to dangerous conditions.

The system provides valuable data for predictive maintenance. By analyzing long-term trends in gas concentrations, operators can identify patterns and potential issues before they become critical, helping to reduce operational disruptions and improve overall infrastructure management. The IoT-based solution is also designed to be scalable and portable, making it adaptable to different confined spaces and applications beyond manholes, such as tunnels, storage tanks, and industrial chambers.

It highlights the critical need for advanced toxic gas detection systems in confined spaces and sets the stage for exploring how IoT technology can be leveraged to enhance safety and operational efficiency in hazardous environments[6].

It sets the stage for a detailed exploration of the IoT-based toxic gas detection and monitoring system, emphasizing its importance in addressing the ongoing safety challenges faced by workers in confined spaces. The rest of this study will explore the design, implementation, and testing of this system, with a focus on its practical application in real-world environments and the potential to save lives.

1.1 Overview

The safety of workers in confined spaces such as manholes is a growing concern due to the hazardous conditions they often face. These environments are prone to the accumulation of toxic gases like methane, hydrogen sulfide, and carbon monoxide, as well as the risk of oxygen deficiency. Prolonged exposure to these gases can cause serious health issues, including respiratory failure, unconsciousness, or even death. The challenge lies in the timely detection and response to these gases to ensure the safety of workers operating in such environments.Traditional gas detection methods, which rely on handheld devices or manual periodic checks, are often inadequate in preventing accidents. These methods are reactive rather than proactive, providing limited real-time monitoring or early warning systems[7]. This gap in safety standards has paved the way for more advanced technologies, specifically those utilizing the Internet of Things (IoT).

IoT-based toxic gas detection and monitoring system is designed to tackle these challenges by offering continuous, real-time surveillance of gas levels inside manholes and similar confined spaces[2][8][1]. Using a network of gas sensors, the system can detect the presence of harmful gases and abnormal oxygen levels, transmitting this information to a central platform via wireless communication technologies like GSM or Wi-Fi[9][10][11]. The data is then processed and displayed on cloud-based systems, which can be accessed remotely through mobile devices or web interfaces. This system offers numerous advantages over traditional methods. It provides workers and supervisors with instant alerts if gas levels exceed safe thresholds, allowing for immediate evacuation and response. Additionally, it enables remote monitoring, eliminating the need for constant physical checks, thus improving

Figure 1.1: Manhole

both efficiency and safety. The IoT platform also facilitates data logging, which can be used for trend analysis, enabling predictive maintenance and preventive measures.

The IoT-based system improves worker safety by continuously monitoring hazardous gas levels, providing early warnings, and allowing for proactive risk management[12]. Moreover, it opens new possibilities for efficient infrastructure management by offering insights into environmental conditions inside man is a immediate safety concern and long-term operational efficiency, making it a valuable solution for confined space safety management. Manholes, sewers, and other confined spaces play a vital role in urban infrastructure, providing access to essential utility systems like sewage, drainage, telecommunications, and electrical networks. These spaces, however, are often hazardous environments for the workers who must enter them for routine maintenance, repairs, or inspections. One of the most significant risks they face is exposure to toxic gases, including methane, hydrogen sulfide, carbon monoxide, and oxygen-deficient atmospheres. These gases, being invisible and often odorless, can accumulate without notice, leading to dangerous or even fatal conditions for workers[13][14].

Despite the known dangers, many current safety practices rely on manual gas detection methods using handheld sensors. While useful, these devices require constant monitoring and are subject to human error. They may fail to provide timely alerts if workers are already inside a manhole when gas levels suddenly rise to hazardous concentrations. As a result,

many accidents occur due to a lack of real-time data and warnings. Thus, the need for a more advanced, automated system to detect and monitor toxic gases is evident, especially for confined spaces like manholes where traditional safety measures are insufficient[15]. The Role of IoT in Safety Systems are The Internet of Things has emerged as a transformative technology across many industries, providing interconnected networks of sensors, devices, and software to enable real-time monitoring, data collection, and control[7]. In the context of confined space safety, IoT-based systems have the potential to revolutionize how gas detection is handled. By deploying a network of gas sensors connected to IoT devices, it becomes possible to continuously monitor gas levels in manholes and provide immediate warnings to
workers and supervisors when dangerous conditions arise.

1.2 Motivation behind the work

The motivation for developing an IoT-based toxic gas detection and monitoring system for manholes and workers stems from the urgent need to enhance safety in hazardous work environments and mitigate the risks associated with exposure to toxic gases. Each year, numerous accidents occur in confined spaces due to insufficient monitoring and detection of harmful gases, resulting in severe injuries and fatalities among workers. These incidents are often exacerbated by the lack of real-time awareness of the environmental conditions inside manholes and similar spaces. Traditional methods of gas detection rely heavily on manual checks and portable devices, which can be ineffective, especially in emergency situations
where immediate action is required.

Furthermore, as urbanization continues to expand, the complexity and frequency of maintenance work in underground infrastructures increase, necessitating more robust safety protocols. The inherent risks involved in these tasks underline the importance of implementing innovative technologies that can provide continuous, reliable monitoring of air quality. IoT technology presents a transformative opportunity to revolutionize safety practices in these environments. By harnessing the power of connected sensors and real-time data transmission, the proposed system aims to provide a comprehensive solution that not only detects hazardous gases but also ensures that timely alerts are sent to workers and supervisors, thereby allowing for prompt evacuation and reducing the likelihood of accidents.

The Figure 1.2 shows the number of deaths of people died by involving in the manhole. By integrating IoT technology into gas detection systems, companies can demonstrate their commitment to worker safety, comply with safety regulations, and potentially reduce liability and insurance costs associated with workplace accidents. The use of a proactive monitoring system not only safeguards the health of workers but also promotes a culture of safety, where employees feel valued and protected. This motivation to prioritize worker safety, combined with the potential for technological advancement in monitoring hazardous environments, drives the urgency to develop an effective and scalable IoT-based toxic gas detection system. Ultimately, the goal is to create a safer work environment that minimizes risks and enhances operational efficiency, paving the way for safer urban infrastructure management.

The advent of IoT technology offers unprecedented opportunities to revolutionize safety in these environments. By integrating real-time gas monitoring with smart communication capabilities, workers can be alerted instantly to dangerous gas concentrations, significantly reducing response times and minimizing health risks. The data collected from these systems can provide valuable insights into gas behavior, informing maintenance practices and regulatory

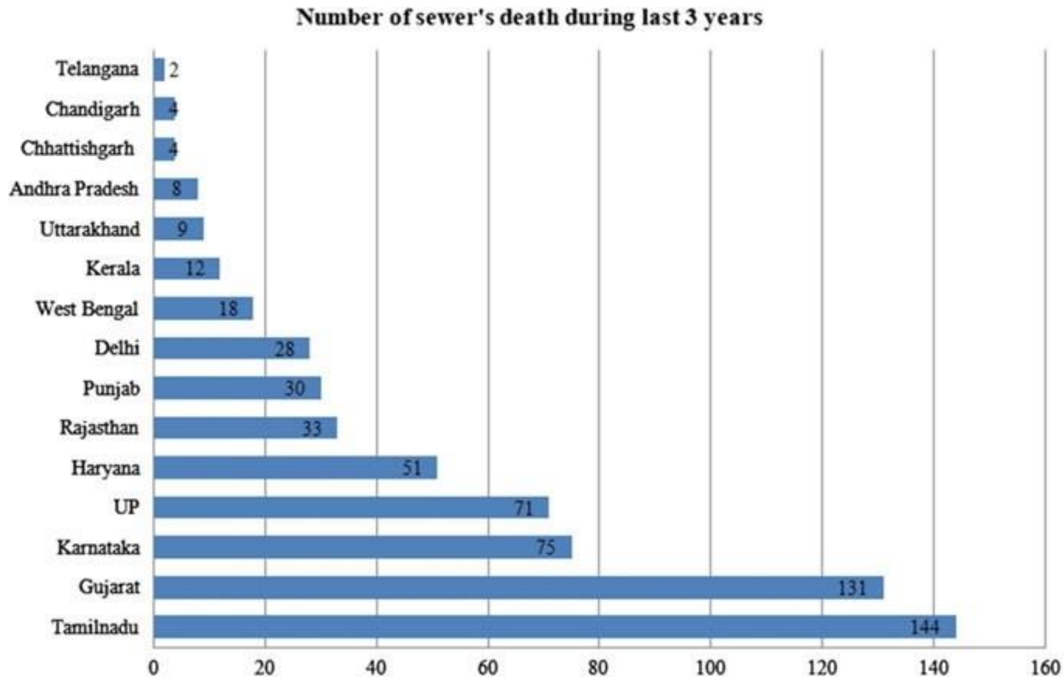

Figure 1.2: Death rate of People dying due to Manhole in India

compliance. This proactive approach not only protects the health and safety of workers but also contributes to the overall efficiency of operations, as organizations can address potential issues before they escalate into critical situations.

Moreover, the increasing urbanization and aging infrastructure of many cities further amplify the urgency of this work. As more workers are deployed to manage and maintain underground utilities, ensuring their safety through innovative technologies becomes paramount. By leveraging IoT solutions, this project aims to foster a culture of safety, where technology acts as a partner in safeguarding human lives, and provides an adaptable framework that can be scaled and applied to various confined spaces. Ultimately, the motivation lies in the belief that every worker deserves to return home safely, and this system serves as a vital step toward achieving that goal.

CHAPTER 2

LITERATURESURVEY

2.1 Introduction

A manhole is an entry point that provides access to underground utility spaces such as sewer systems, stormwater drainage, electrical ducts, and other essential infrastructure. These access points are typically located on streets, sidewalks, or other surfaces, marked by a circular or square cover. Manholes play a crucial role in enabling maintenance and inspection of these hidden utilities, as well as in carrying out repairs or emergency interventions when required[16]. Regular cleaning and maintenance of manholes are essential to prevent blockages, which could otherwise disrupt the flow of waste and water and lead to serious issues such as flooding, system malfunctions, or environmental contamination.

The process of cleaning a manhole involves clearing out debris, sludge, silt, and other accumulated materials that can obstruct the pipes and hinder smooth operations. Over time, various types of waste accumulate within these systems, potentially clogging the pipes and causing backups that can affect nearby neighborhoods. This waste removal process requires specialized equipment, including high-pressure water jetting machines to dislodge stubborn sludge and vacuum or suction machines to extract the debris efficiently. In addition to machinery, technicians also use protective gear and follow strict safety protocols, as manholes may contain hazardous gases, such as methane or hydrogen sulfide, that pose risks to workers.

Maintaining a regular schedule for manhole cleaning is critical. When left unattended, blockages can form, leading to system breakdowns or even environmental hazards as untreated wastewater seeps into surrounding soil and water bodies. This proactive approach also extends the lifespan of sewer and drainage systems, reducing the need for costly repairs. Clean and well-maintained manholes thus contribute to the safety, efficiency, and sustainability of urban infrastructure. In sum, routine manhole maintenance is a vital aspect of public health and environmental protection, ensuring that urban drainage and waste management systems operate smoothly and effectively.

2.2 Health Monitoring Technologies

Health monitoring technologies have evolved significantly, particularly in applications for high-risk environments where real-time data on workers' health conditions is crucial. In confined spaces such as manholes, these technologies are essential for detecting signs of physical stress, fatigue, and potential health crises before they escalate asshown in Figure 2.1. Various sensor-based solutions enable continuous monitoring of vital signs such as heart rate, body temperature, blood oxygen levels, and even respiratory rate. Photoplethysmography

Figure 2.1: Worker Cleaning the Mnahole

and Electrocardiography sensors are widely used to assess cardiovascular health, measuring parameters like heart rate and rhythm to detect abnormalities. These sensors can provide early warnings of stress or cardiovascular strain, helping to prevent incidents caused by sudden health deterioration. Other commonly used sensors include temperature and respiratory monitoring devices, which track body temperature and respiratory rate both critical indicators in environments where exposure to gases or lack of oxygen is possible[17].

2.2.1 MEMS Technology

Micro-Electro-Mechanical Systems technology plays a pivotal role in health monitoring applications, offering compact and efficient solutions for tracking physiological parameters. MEMS devices are integrated with various sensors capable of measuring vital signs such as heart rate, blood pressure, respiratory rate, and body temperature. These miniature sensors in the Figure 2.2 utilize different principles, including capacitive, resistive, and piezoelectric methods, to detect changes in physical properties associated with health metrics. MEMS accelerometers can be employed to monitor movement and activity levels, while MEMS-based gyroscopes aid in balance and orientation assessments. In wearable health devices, MEMS sensors provide continuous monitoring, facilitating the early detection of health issues and enabling timely interventions[18].

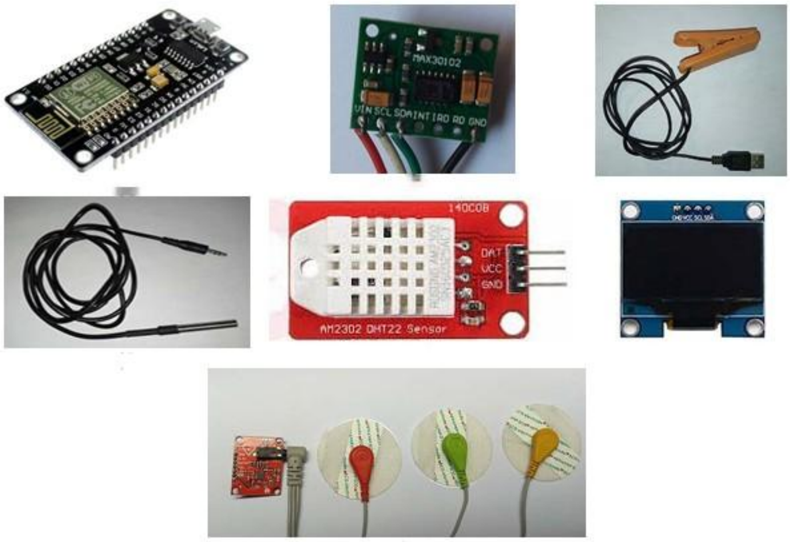

Figure 2.2: Various Sensor for Health Monitoring

Moreover, MEMS technology enables the development of sophisticated biosensors that can detect biomarkers in bodily fluids, providing insights into metabolic processes and disease states. These biosensors often incorporate nanostructured materials to enhance sensitivity and specificity for targeted analytes.

Integration with wireless communication technologies allows for real-time data transmission to mobile devices or cloud platforms, empowering patients and healthcare providers with accessible health data. This connectivity supports telemedicine applications, remote patient monitoring, and personalized healthcare, ultimately enhancing patient outcomes and promoting proactive health management. Overall, MEMS technology is transforming health monitoring by making it more efficient, accurate, and user-friendly, paving the way for innovative healthcare solutions.

2.2.2 PPG and ECG Monitoring Systems

Photoplethysmography and Electrocardiography sensors have become essential tools for real-time monitoring of heart rate and cardiovascular health, especially in high-risk environments like confined spaces[10][12]. In Figure 2.4 PPG signals are obtained by a sensor that operate by emitting light into the skin and measuring changes in light absorption caused by blood flow, providing a non-invasive way to monitor heart rate, blood volume, and oxygen saturation. PPG sensors are particularly useful in wearables because they are compact, energy-efficient, and capable of continuous monitoring, which is ideal for tracking worker health in dynamic, confined environments[19][11].

ECG sensors, on the other hand, directly measure the electrical activity of the heart, offering more detailed data on heart rate and rhythm. ECG readings from the Figure 2.3 provide critical insights into cardiovascular health, detecting abnormalities like arrhythmias,

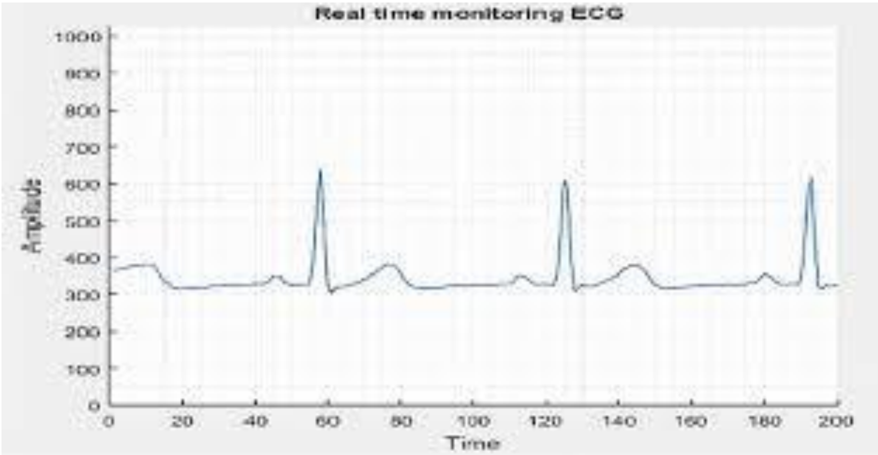

Figure 2.3: ECG Monitoring Graph of a Person

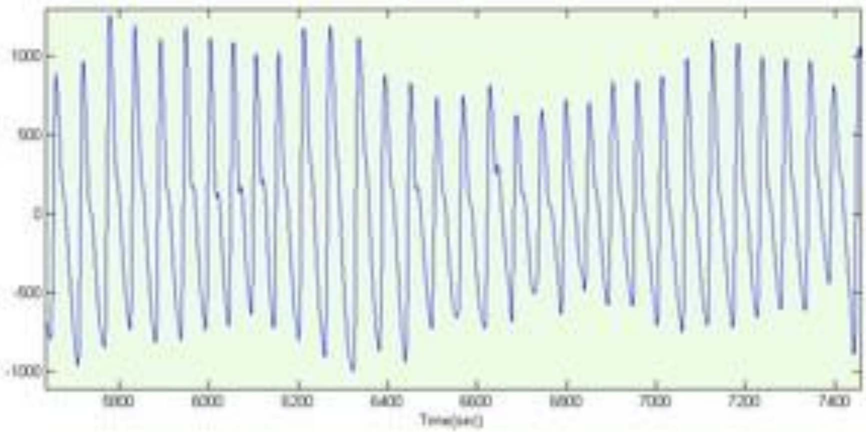

Figure 2.4: PPG Monitoring Graph of a Person

which may indicate stress or an imminent cardiac event. ECG is often considered a gold standard in medical heart monitoring due to its accuracy and detailed output[20][21]. However, it requires contact with the skin through electrodes, making it slightly more intrusive than PPG[22].

Several studies have demonstrated the application of PPG and ECG sensors for worker health monitoring in confined or hazardous spaces. For instance, research on wearable ECG devices for mine workers has shown that these sensors can reliably detect early signs of cardiovascular distress, prompting timely intervention to prevent incidents. In another study focused on industrial health monitoring, PPG sensors were used to monitor heart rate and oxygen saturation in workers exposed to high temperatures and physical exertion, providing early indicators of heat stress and cardiovascular strain.

The combination of PPG and ECG sensors in wearable devices offers a comprehensive solution for monitoring cardiovascular health in real-time. Such integration enables early detection of stress-related anomalies, which is especially beneficial in confined spaces where immediate medical assistance may be limited. In environments like manholes, where exposure to toxic gases or low oxygen levels can exacerbate cardiovascular issues, these sensors play a critical role in safeguarding workers by continuously relaying health data to safety personnel. This approach has become integral to improving worker safety and reducing the incidence of health related accidents in confined, high-risk environments.

2.2.3 Temperature and Respiratory Monitoring Systems

Temperature and respiratory monitoring play crucial roles in ensuring worker safety, especially in confined spaces where environmental hazards and physical strain can pose significant health risks. Temperature sensors, such as the DHT11, are commonly used to monitor a worker’s core body temperature as well as ambient conditions in the environment[10]. The DS18B20, in particular, can provide readings of both temperature and humidity, allowing a comprehensive assessment of the conditions within a confined space[23]. Monitoring temperature is essential in these environments since high temperatures can quickly lead to heat exhaustion or heatstroke, while high humidity levels can exacerbate respiratory stress. By tracking these metrics, systems can detect early signs of heat stress, enabling timely interventions to prevent heat-related illnesses.

SPO2 sensors measure oxygen saturation in the blood, providing an indication of a worker’s respiratory health and overall oxygen levels. Oxygen saturation is particularly critical in confined spaces, where workers are at risk of exposure to gases like methane or carbon monoxide, which can displace oxygen. Low oxygen levels can lead to symptoms like dizziness, confusion, or even loss of consciousness, posing immediate threats to safety. By using AD8232 ECG sensors, the system can continuously monitor blood oxygen levels, detecting hypoxia before it becomes life threatening.

In confined or poorly ventilated spaces, toxic gases and reduced oxygen levels create an invisible but severe risk to respiratory health. The combined use of temperature and SPO2 sensors in IoT-enabled monitoring systems can therefore provide real-time, actionable data on a worker’s physical and respiratory condition. Continuous temperature and respiratory monitoring allow the system to flag abnormal readings, alerting personnel to potential health issues related to high temperatures, humidity, or low oxygen levels. Together, these sensors contribute to a proactive approach to worker safety, preventing incidents and ensuring timely medical responses in high-risk environments.

2.2.4 Wearable IoT Systems

Wearable IoT systems for health monitoring have transformed workplace safety by providing continuous, real-time insights into workers' physical conditions, particularly in high risk environments such as confined spaces[24]. These wearable devices integrate various sensors, including heart rate, temperature, and SPO2 sensors, to create a comprehensive health

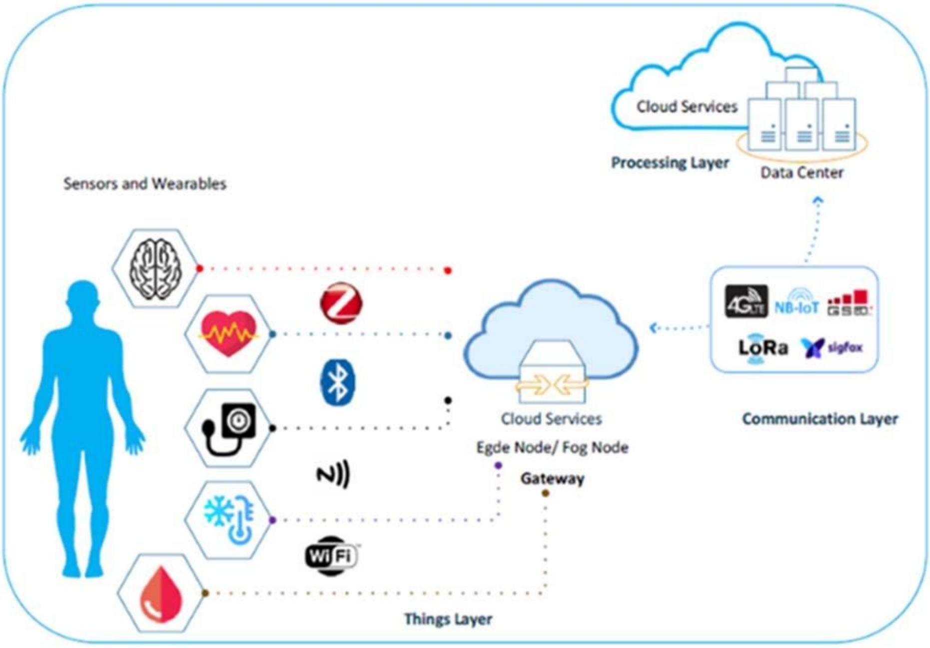

Figure 2.5: Wearable Sensors Connected to the IoT

profile of the worker. By combining these metrics, wearable devices offer a detailed overview of an individual's cardiovascular health, respiratory condition, and susceptibility to physical stressors.

For heart rate monitoring, wearable devices typically employ PPG or ECG sensors, which provide accurate measurements of heart rate and detect irregularities that may indicate stress or fatigue. Temperature sensors, like the DHT11, monitor both the worker's body temperature and the ambient environment, offering critical data in settings prone to heat stress. The SPO2 sensor measures oxygen saturation levels, providing real-time feedback on respiratory health, which is vital in confined spaces where gas exposure and oxygen deficiency are major concerns. Wearable IoT devices in Figure 2.5, equipped with wireless connectivity, transmit this health data to centralized systems, allowing supervisors to monitor multiple workers simultaneously and intervene in emergencies. Alerts can be triggered if any of the health metrics reach dangerous thresholds, prompting rapid responses to safeguard worker health. Many wearables also include GPS functionality, which enables location tracking for quicker access during emergencies.

2.3 Toxic Gas Detection and Environmental Monitoring

Detecting and visualizing the dangerous area of leaking toxic gases is important for large-scale petrochemical plants[13][14][25][5]. Thereare many kinds of toxic gases in petrochemical plants, e.g., sulfuretted hydrogen, hydrogen chloride, and sulfur dioxide. Once these toxic gases leak, they can cause an explosion that results in serious economic loss. For example, if the SINOPEC Maoming Petrochemical Company stops producing for one day, the economic loss is almost 10,000,000 RMB.1 Explosions also seriously threaten the lives and safety of local citizens who are living around the leakage. To avoid this kind of explosion, and reduce the toxic gas threat to local citizens, many methods have been proposed to detect and visualize the concentration of toxic gases. These methods use special monitoring devices to provide gas concentration reports within their individual ranges[26]. Because of the continuity of gas diffusion and the invisibility of toxic gases, it is difficult to detect and visualize such a continuous object using only the scattered concentration reports.

2.3.1 Nanostructured Metal Oxide Semiconductor for Gas Detection

Nanostructured metal oxide semiconductor sensors have emerged as a promising technology for enhancing sensitivity in toxic gas detection within environmental settings. These sensors leverage the unique properties of nanomaterials, which exhibit a high surface area-to-volume ratio, allowing for greater interaction with gas molecules. The nanoscale dimensions of the metal oxide materials, such as tin oxide, zinc oxide, and titanium dioxide, enhance their electronic properties and catalytic activity, resulting in improved sensitivity and faster response times to various toxic gases[27].

When exposed to target gases, the nanostructured MOS sensors undergo a change in resistance due to the adsorption of gas molecules on their surface. This change can be quantitatively measured, allowing for the detection and monitoring of harmful gases like carbon monoxide, nitrogen dioxide, and volatile organic compounds at low concentrations. The incorporation of nanostructures can further optimize sensor performance through techniques like doping, which can modify the electrical properties of the metal oxides, and the use of composite materials, which can enhance selectivity and stability. The compact size and low power requirements of nanostructured sensors make them ideal for integration into portable and distributed sensing networks. These networks can provide real-time monitoring in various environmental contexts, including urban areas, industrial sites, and natural ecosystems, contributing to better air quality management and public safety. Moreover, advancements in nanofabrication techniques and sensor design continue to push the boundaries of sensitivity, selectivity, and reliability, making nanostructured metal oxide semiconductor sensors a key component in the future of environmental monitoring and toxic gas detection technologies.

2.3.2 Gas Monitoring Systems

Gas detection in confined spaces, such as manholes, is crucial for ensuring the safety of workers who may be exposed to toxic or flammable gases. Among the commonly used sensors, the MQ4 sensor is designed for detecting methane. This sensor operates on the principle of metal oxide semiconductor technology, which measures changes in resistance due to the presence of methane. It has a detection range of 300 to 10,000 ppm and is widely

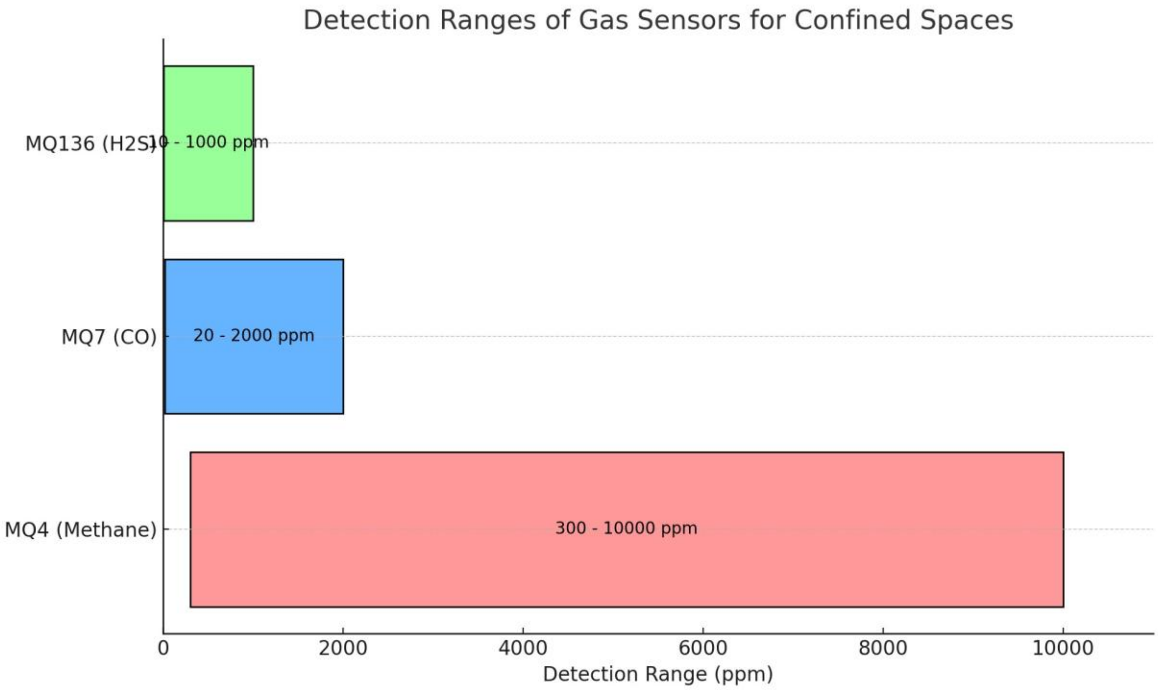

Figure 2.6: Gas Sensors Detection Ranges

utilized for monitoring gas leaks in residential and industrial applications. The MQ4 sensor is sensitive to humidity and temperature fluctuations, which can affect its accuracy, and it requires periodic calibration to maintain optimal performance[27][28][29].

Another important sensor is the MQ7, which detects carbon monoxide. Like the MQ4, it utilizes a similar resistive sensing mechanism. The MQ7 has a detection range of 20 to 2000 ppm and is commonly employed in fire detection systems and vehicle emissions monitoring. Despite its effectiveness, this sensor can also be influenced by other gases, leading to cross-sensitivity issues, and it too needs regular calibration to ensure accurate readings, particularly in varying environmental conditions[30][31]. The MQ136 sensor is specifically designed for the detection of hydrogen sulfide. This sensor functions similarly to the MQ4 and MQ7, changing resistance in response to HS concentrations. With a detection range of 10 to 1000 ppm, the MQ136 is essential in environments such as sewage systems and petrochemical industries, where HS exposure poses significant health risks. As with the other sensors, environmental factors can impact its performance, and calibration is necessary to maintain accuracy. The Figure 2.6 shows the various detection ranges of different sensors[32].

2.3.3 Colorimetric Array Gas Detection

A colorimetric array with a microcontroller is an advanced solution for detecting various gases in real-time by utilizing a combination of chemical sensors and digital processing capabilities. This system employs an array of colorimetric sensors, each designed to respond to specific gases by changing color in reaction to their presence. The sensors are typically coated with specific reagents that interact with target gases, leading to a measurable color change proportional to the concentration of the gas[33].

The microcontroller acts as the central processing unit, interpreting the signals from the sensors. It uses analog-to-digital conversion to translate the colorimetric changes into quantifiable data, allowing for precise gas concentration measurements. The microcontroller can also handle multiple sensor readings simultaneously, making it capable of detecting and distinguishing between various gases, such as carbon dioxide, methane, and volatile organic compounds.

This system can be enhanced with connectivity features, allowing it to transmit data to remote monitoring stations or mobile devices for real-time analysis. With integrated algorithms, the microcontroller can perform data analysis, calibration, and even machine learning tasks to improve the accuracy and reliability of gas detection. The combination of colorimetric sensing and microcontroller technology provides a portable, efficient, and cost-effective method for environmental monitoring, industrial safety, and personal safety applications. By continuously analyzing the air quality, such systems play a vital role in ensuring safe living and working conditions while facilitating quick responses to potential gas leaks or hazardous situations.

2.3.4 Air Quality and Ventilation Systems

Research on air quality and ventilation systems in confined spaces has increasingly focused on advanced purification and neutralization technologies, particularly in applications where pollutant levels need continuous monitoring and rapid mitigation. Technologies such as activated carbon filters in Figure 2.7, filters, and UV-C light purifiers are widely used to remove or neutralize particulates, pathogens, and volatile organic compounds from the air. In industrial and laboratory settings, electrostatic precipitators and ionizers also help to reduce airborne contaminants effectively.

IoT-enabled air quality systems have emerged as a key advancement, leveraging sensors to detect hazardous gases like carbon monoxide, methane, and ammonia. These sensors relay real-time data to a centralized system, which can trigger alerts or directly engage ventilation systems shown in Figure 2.8 when toxic levels are detected. For instance, some IoT systems activate exhaust fans, open windows, or release neutralizing agents automatically to restore safe air conditions. Integrating AI-based analytics with these systems allows predictive maintenance and adjustments based on usage patterns, enhancing both air quality and operational efficiency. To improve safety in high-risk areas, toxic gas sensors can trigger alarms or activate exhaust fans, thus reducing exposure risks. Additionally, IoT-based analytics allow for predictive maintenance by alerting facility managers when filters need changing or other maintenance actions are required, thus ensuring sustained air quality[34][18]. However, these IoT-based solutions also face challenges. In underground or enclosed environments

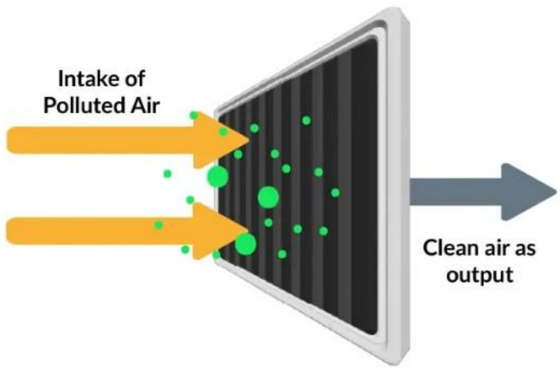

Figure 2.7: Carbon Filters

Figure 2.8: Air Ventilation Systems

with weak connectivity, data transmission can be difficult, affecting the real-time response capability. For these cases, low-power wide-area networks like LoRa or mesh networks are sometimes used as alternatives to GSM-based communication, ensuring data can still be transmitted efficiently. Despite these limitations, IoT-enhanced air quality management offers a proactive approach to maintaining healthy, breathable air in confined or high-risk environments, making workplaces and living spaces safer and more comfortable.

2.4 IoT-Based Communication and Alert Systems

2.4.1 LoRa Communication Technologies

LoRa (Long Range) technology is a low-power, wide-area networking protocol designed for the Internet of Things, enabling long-range communication between devices while consuming minimal power. It allows for data transmission over distances of up to 15 kilometers in rural areas and several kilometers in urban settings, making it ideal for applications requiring extensive coverage. LoRa devices, equipped with transceivers, send small, infrequent data packets, making them suitable for applications like temperature and gas detection[35][36].

The technology consists of LoRa devices, gateways that connect these devices to the internet, and servers that manage data processing. Its robustness enables reliable communication even in challenging environments, supporting various applications such as smart agriculture, smart cities, environmental monitoring, asset tracking, industrial IoT, and healthcare. The combination of long range capabilities, low power consumption, and scalability makes LoRa particularly advantageous for IoT deployments where traditional networking solutions may not be feasible. LoRa technology operates in the sub-gigahertz frequency bands, such as 433 MHz, 868 MHz, and 915 MHz, allowing it to penetrate obstacles like buildings and foliage effectively. Its modulation technique, known as Chirp Spread

Spectrum, enhances the signal's robustness against interference and allows for longer communication ranges compared to conventional radio technologies.

2.4.2 Data Processing and Cloud Integration

Cloud-based IoT data processing has become essential in monitoring and analyzing long-term health trends, especially in fields like occupational health and environmental safety[37][36][35]. IoT devices continuously collect data on air quality, temperature, noise levels, and biometric markers, which can be stored and processed in the cloud for real-time insights and comprehensive analytics.

Cloud integration enables real-time monitoring, where alerts are triggered if conditions surpass safe limits, such as high pollutant levels or elevated temperatures. This capability is critical in high-risk environments like factories, mines, or construction sites, ensuring swift responses to protect workers' health. Additionally, cloud storage supports the aggregation of long-term data, enabling trend analysis and predictive insights. Advanced analytics powered by AI can identify patterns that inform preventive measures, helping to mitigate health risks over time and providing valuable insights into worker well-being on a large scale.

By using cloud-based IoT platforms, organizations can enhance workplace safety and health management with real-time updates and long-term health monitoring in one unified system[38][39][16].

2.4.3 ZigBee Communication

ZigBee technology is a wireless communication protocol specifically designed for low-power, low-data-rate applications, making it ideal for automation and monitoring systems. Operating on the IEEE 802.15.4 standard, ZigBee is known for its energy efficiency and robust networking capabilities, allowing battery-operated devices to function for extended periods without needing replacements[40]. Its support for mesh networking enables devices to communicate directly with one another, relaying messages over multiple hops, which enhances the network's range and reliability. ZigBee is also cost-effective, with inexpensive components that make it suitable for a wide range of applications, including home automation, industrial control systems, and environmental monitoring. With data rates ranging from 20 to 250 kbps, it is capable of supporting up to 65,000 devices in a single network, offering scalability for various needs. This technology is commonly used in smart home devices like lighting controls and security systems, as well as in industrial settings for process control and equipment monitoring. the technology's interoperability with other standards and devices enhances its applicability in diverse environments, such as smart cities and industrial IoT (Internet of Things) solutions. ZigBee's ability to seamlessly integrate with gas sensors makes it a valuable option for hazardous gas detection systems.

2.5 Conclusion

The critical role of IoT-based monitoring and alerting systems in enhancing safety in confined spaces[41]. Various studies emphasize the importance of real-time data collection, remote monitoring, and automated alert mechanisms, especially in environments where exposure to hazardous conditionssuch as toxic gases, extreme temperatures, or low oxygen levels poses significant risks to worker safety. Technologies like GSM modules for communication, cloud platforms for data processing and storage, and mobile applications for instant notifications are commonly employed to ensure swift responses during emergencies[42]. The literature reveals the increasing trend of integrating cloud analytics with IoT systems to track long-term health trends and environmental conditions. This integration provides valuable insights for predictive maintenance and risk assessment, helping to reduce incident rates over time. However, challenges related to connectivity, especially in remote and underground locations, are frequently noted.

The survey underscores the advantages of combining IoT, cloud computing, and automated alert mechanisms to create a proactive approach to safety management in confined spaces. By bridging real-time monitoring with advanced analytics, IoT-enabled systems offer not only immediate risk mitigation but also the capacity for ongoing health and safety improvements in high risk environments. This knowledge base informed the design and development of our project, aligning it with best practices and addressing common challenges identified in existing research.

CHAPTER 3

METHODOLOGY

3.1 Introduction

The methodology for designing and implementing the IoT-based manhole maintenance worker monitoring and toxic gas detection system follows a structured, step-by-step approach. The process begins with a requirement analysis, where the project objectives are clearly defined. The primary goal is to develop a system that continuously monitors the health of workers within manholes and detects hazardous gases, providing real-time alerts to relevant authorities. During this phase, consultations with key stakeholders, such as municipal authorities, safety experts, and manhole workers, help identify critical safety parameters, including the vital health signs to be monitored (e.g., heart rate, oxygen saturation) and the types of gases to be detected (e.g., methane, carbon monoxide, hydrogen sulfide). These requirements serve as the foundation for the system design.

In the system design and architecture phase, the hardware and software components of the system are selected and integrated. The hardware includes various IoT sensors for measuring worker health and environmental conditions. Wearable devices are designed to capture real-time data on vital signs, while gas sensors are used to detect dangerous concentrations of toxic gases in the manhole. A microcontroller, such as an Arduino or Raspberry Pi, serves as the central processing unit, collecting data from all sensors. This phase also involves selecting communication protocols to ensure that data is transmitted in real-time to a central monitoring system, even in remote or underground locations with limited network coverage. The software architecture is designed to process sensor data, trigger alerts when necessary, and provide visualization for easy monitoring.

During the system development phase, the sensors and communication modules are integrated into the IoT framework. Wearable health sensors are connected to the worker via comfortable, non-intrusive devices that can be worn during operations. Toxic gas sensors are installed within the manhole to detect hazardous gases like methane, carbon monoxide , and hydrogen sulfide. The microcontroller receives the sensor data and processes it in real-time. If any abnormal readings are detected, such as a worker's heart rate going outside safe limits or dangerous gas levels being detected, the system is programmed to send immediate alerts to the authorities via mobile networks, cloud platforms, or dedicated applications.

The next step is data transmission and alert mechanism implementation. Real-time data collected from the sensors is transmitted wirelessly to a centralized server or cloud-based platform. Here, data processing algorithms analyze the incoming data to detect any irregularities.

Threshold values for health parameters and toxic gas levels are established based on safety standards, and the system is configured to send automatic alerts to relevant personnel (e.g., municipal authorities, emergency responders) when critical levels are reached. The alert mechanism may include SMS, email notifications, or direct app-based alerts, ensuring that any hazard is communicated immediately to enable rapid response.

In the testing and validation phase, the system is subjected to extensive testing in real-world conditions to ensure its accuracy, reliability, and durability. The wearable devices are tested on workers to validate the health monitoring capabilities, ensuring that data such as heart rate, oxygen levels, and temperature are accurately captured and transmitted in real-time. The gas sensors are tested for sensitivity and accuracy in detecting hazardous gases within confined spaces like manholes. Stress testing is performed under different environmental conditions to ensure the system's robustness, especially in environments with fluctuating temperatures or humidity levels. The data transmission and alert systems are tested to verify that alerts are sent in a timely manner, with minimal delays, and received by the correct authorities.

Finally, during the deployment and maintenance phase, the system is deployed across various manhole locations. Continuous monitoring of worker health and manhole environments is maintained, with periodic updates and maintenance checks to ensure long-term reliability. Feedback from end-users (both workers and authorities) is collected to improve the system and address any challenges faced during actual operations. The system is designed for scalability, allowing for the addition of more sensors or manhole locations as needed, and it is continuously updated to incorporate new safety regulations and technological advancements. This methodology ensures that the IoT-based system is designed to be practical, efficient, and highly responsive, offering a robust solution for improving worker safety in hazardous manhole environments while enabling quick and effective emergency response.

3.2 Block Diagram

This project focuses on designing and implementing a comprehensive IoT-based system to enhance the safety of manhole maintenance workers and monitor environmental conditions within confined spaces. The first component of the system is a Manhole Gas Monitoring and Neutralization System, which is designed to continuously track environmental parameters such as temperature, oxygen levels, and the presence of toxic gases like methane or carbon monoxide. Sensors feed this data into a central

Microcontroller, which processes the information and, if hazardous conditions are detected, activates a motorized neutralization mechanism. The system also sends alerts to authorities via a GSM Module and provides real-time data to a Cloud Platform for remote monitoring. In addition to environmental monitoring, the project incorporates a Worker Health Monitoring System to track the vital signs of maintenance workers in real time. The system includes sensors that monitor body temperature, oxygen saturation (SPO2), and heart rate. The microcontroller processes this health data, sending alerts if any of these vital parameters exceed safe limits. Similar to the gas monitoring system, the health data is transmitted via GSM to authorities, while a cloud-based platform allows for continuous remote monitoring of worker health, improving the response time in case of emergencies. Finally, the system includes a Manhole Condition Monitoring System, which assesses the structural and environmental integrity of manholes. A Tilt Sensor monitors any shifts or structural damage, while a Water Level Sensor detects flooding or water ingress. The alert mechanism encompasses both SMS notifications sent through the GSM module and alerts displayed on the cloud interface, ensuring that relevant personnel are promptly informed of any safety issues. The entire system operates on a suitable power supply, facilitating the effective functioning of all components. Overall, the block diagram effectively communicates the flow of data and interactions between each element, highlighting how the integration of various technologies creates a comprehensive safety monitoring solution for confined spaces, enhancing both real-time responses and long-term health tracking. This data is processed by the microcontroller and relayed to the authorities and the cloud platform in real time. Together, these interconnected systems create a robust safety framework for manhole workers, providing real-time monitoring, early warnings, and rapid response capabilities, significantly reducing the risks associated with confined-space work.

3.2.1 Block Diagram of Manhole Monitoring System

The Manhole Gas Monitoring and Neutralization System is an IoT-based solution designed to ensure the safety of workers operating in hazardous environments by continuously monitoring and managing toxic gas levels within manholes. At the heart of this system is a Microcontroller that gathers data from various sensors strategically placed within the manhole. These sensors include a Gas Sensor that detects the presence of harmful gases such as methane, carbon monoxide, and hydrogen sulfide. In addition, a Temperature Sensor monitors the environmental heat, as high temperatures can indicate dangerous conditions, while an Oxygen Sensor ensures that the oxygen levels remain within safe limits for workers.

When the gas sensor detects elevated levels of toxic gases beyond the pre-defined safety thresholds, the microcontroller triggers a neutralization process. This is carried out by a Motor Driver that activates a Motor, which could be connected to ventilation fans or other neutralizing mechanisms designed to either disperse or reduce the concentration of harmful gases. This active response helps mitigate immediate risks, providing a real-time solution to gas-related hazards. In Figure 3.1, the system uses a GSM Module to send instant notifications to the relevant authorities or monitoring personnel, alerting them to potential dangers such as toxic gas leaks or critical environmental conditions. These notifications allow for quick decision-making and the dispatch of emergency response teams if needed. Additionally, the system is integrated with a Cloud Platform, where all data from the sensors is uploaded

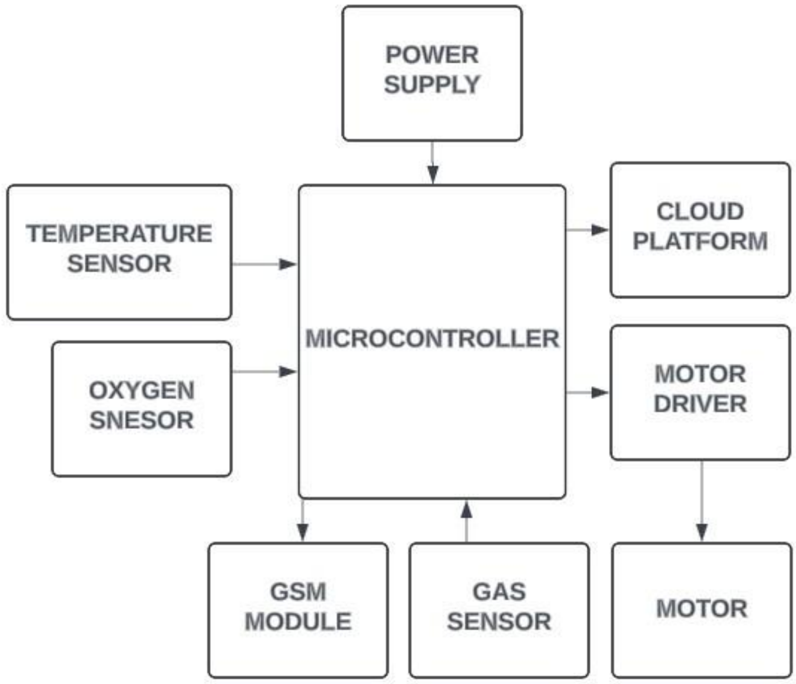

Figure 3.1: Manhole Monitoring System Block Diagram

in real time. This platform enables remote monitoring of multiple manhole sites, allowing authorities to track the conditions without being physically present. The cloud also provides a historical record of data for analysis, helping in preventive maintenance and planning. Overall, the Manhole Gas Monitoring and Neutralization System offers a proactive approach to safeguarding workers from toxic gas exposure, combining real-time monitoring, automatic response mechanisms, and remote communication to prevent accidents and improve response times.

3.2.2 Block Diagram of Manhole Maintainance Worker Monitoring System

The Worker Health Monitoring System is an advanced IoT-based solution designed to continuously monitor the health and vital signs of workers in hazardous environments, such as manholes, to ensure their safety. The block diagram in Figure 3.3 represents core of the system is a Microcontroller that receives data from multiple health related sensors, specifically chosen to track critical indicators like body temperature, heart rate, and blood oxygen levels. The Temperature Sensor is used to measure the worker's body temperature, ensuring that they are not experiencing heat stress or other temperature-related health issues, which can be common in confined spaces. In Figure 3.2, the system includes an SPO2 Sensor, which measures the worker's blood oxygen saturation. This is particularly important in environments where the oxygen levels may drop due to the presence of toxic gases or poor ventilation.

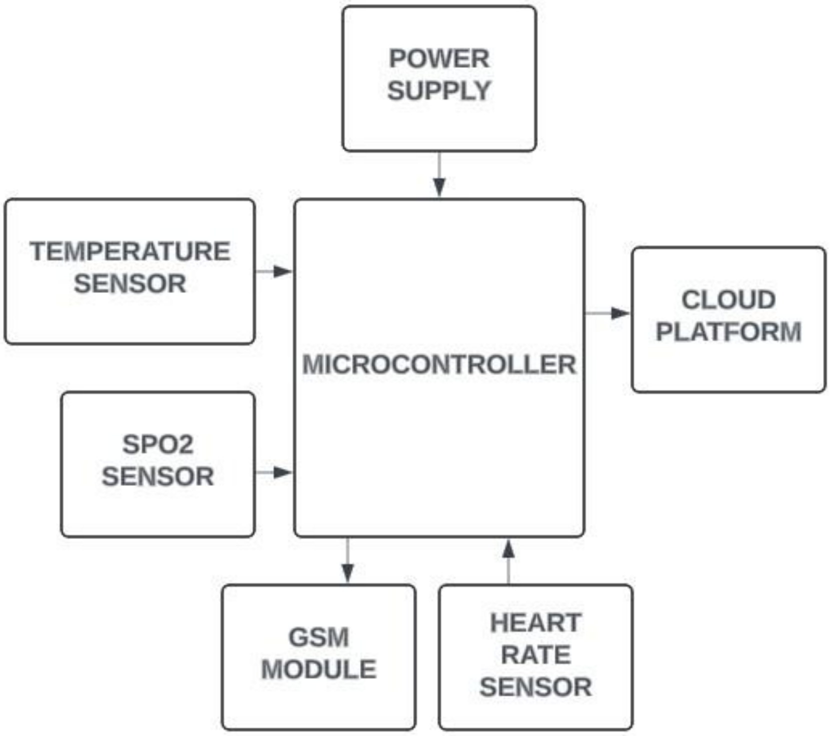

Figure 3.2: Manhole Maintainance Worker Monitoring Sysytem Block Diagram

By continuously monitoring SPO2, the system ensures that workers are not experiencing hypoxia, which can lead to serious health complications. Complementing this is the Heart Rate Sensor, which tracks the worker's pulse in real-time. Abnormal heart rate patterns can indicate stress, physical exhaustion, or other underlying medical conditions that may require immediate attention.

The data collected by the sensors is processed by the microcontroller, which continuously analyzes the vital signs against predefined safety thresholds. If any of these health parameters exceed the safe limits such as a dangerously high heart rate, low oxygen saturation, or extreme body temperature the system immediately sends an alert through a GSM Module to the authorities or monitoring personnel. This rapid alert system ensures that medical attention can be dispatched quickly if the worker is in distress, potentially saving lives. Additionally, the system is connected to a Cloud Platform, which provides remote monitoring capabilities. Health data is uploaded in real time, allowing supervisors and health authorities to track the well being of workers without the need for direct, on-site supervision. The cloud system can also store historical health data for each worker, enabling long-term health monitoring, trend analysis, and preventive care planning. By integrating real-time monitoring with rapid communication and remote data management, the Worker Health Monitoring

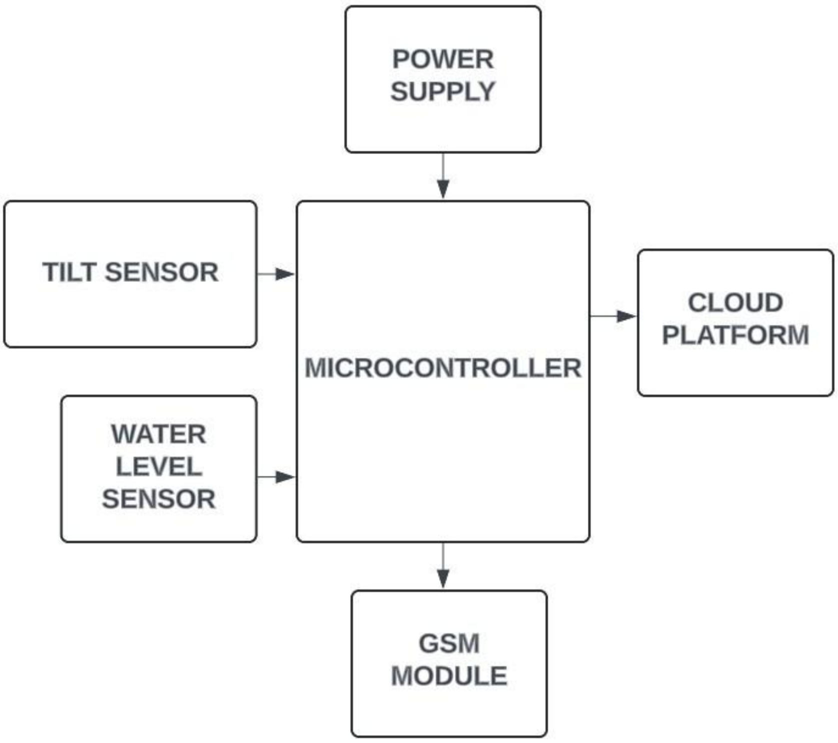

Figure 3.3: Manhole Lid Monitoring System Block Diagram

System significantly improves worker safety in hazardous environments, reducing the risks of accidents and medical emergencies.

3.2.3 Block Diagram of Manhole Lid Monitoring System

The Manhole Condition Monitoring System is a crucial part of ensuring the structural integrity and environmental safety of manholes, especially in hazardous or high-risk areas. This system is designed to continuously monitor key parameters related to the physical condition of the manhole and alert authorities if any unsafe conditions are detected. At its core, the system uses a Microcontroller that collects data from specialized sensors, including a Tilt Sensor and a Water Level Sensor. These sensors are vital for assessing the stability and safety of the manhole, providing real-time insights into any potentially dangerous changes.

The Tilt Sensor plays an essential role in monitoring the structural integrity of the manhole. It detects any unusual shifts, tilts, or movements in the manhole cover or walls, which may indicate structural failure or impending collapse. This is especially important in areas prone to subsidence, heavy vehicle traffic, or seismic activity. Early detection of any movement can trigger preventive maintenance or repairs, thus reducing the likelihood of accidents or manhole failures that could endanger both workers and the general public. The Water Level Sensor monitors the presence of water within the manhole, which is critical in preventing flooding or water ingress, particularly during heavy rains or in flood-prone areas. Elevated water levels can create unsafe working conditions for maintenance workers and may also lead to overflow issues in the sewage or drainage systems. By detecting rising water levels early, this system enables quick action, such as pumping out the water or closing the manhole to prevent accidents. As shown in Figure 3.3 this condition monitoring setup is equipped with a GSM Module for communication, ensuring that alerts are sent immediately to the relevant authorities when abnormal conditions are detected. This facilitates a swift response to potential hazards. Additionally, the system is connected to a Cloud Platform, where all the sensor data is stored and can be accessed remotely. The block diagram in Figure 3.4 shows cloud platform allows for real-time monitoring, data analysis, and record-keeping, enabling authorities to track multiple manholes across a wide area without needing to physically inspect each one. This not only saves time and resources but also allows for a predictive maintenance approach, where repairs and interventions can be scheduled based on data trends.

In summary, the Manhole Condition Monitoring System provides a comprehensive solution to ensure the safety and structural stability of manholes. Through the use of real-time monitoring, automated alerts, and cloud-based data management, this system helps to minimize the risks associated with structural failures and environmental hazards in manholes, ultimately protecting workers and the public.

3.3 Circuit Diagram

The Manhole Gas Monitoring and Neutralization System is designed to ensure the safety of workers by detecting hazardous gases in confined spaces like manholes. The central microcontroller is connected to a gas sensor, which monitors for dangerous gases such as methane or hydrogen sulfide. A temperature and humidity sensor provides additional environmental data. When harmful gases are detected, the system activates a motor through a motor driver to initiate neutralization, possibly through ventilation or chemical dispersion. In the event of danger, the system also uses a GSM module to send real-time alerts to authorities or relevant personnel. The entire setup is powered by a battery, ensuring continuous monitoring even in remote locations.

The Health Monitoring System is intended to keep track of a worker's vital signs while they are working in hazardous environments. This system features sensors that monitor heart rate, blood oxygen levels (SpO2), and body temperature. These vital statistics are processed by the microcontroller, which monitors for any signs of abnormality. In the case of irregular health conditions, the GSM module enables the system to communicate the worker's health status to supervisors or emergency responders, ensuring timely intervention. This system is crucial for maintaining the health and safety of workers in risky environments like manholes.

The Manhole Condition Monitoring System focuses on the physical conditions inside a manhole, such as water levels and potential structural shifts. It incorporates a water level sensor and a tilt sensor to detect changes in the manhole's stability. This system is also powered by a battery, and the data collected is processed by the microcontroller. The tilt sensor ensures that any shifts or movements within the manhole, possibly indicating structural failure, are detected early. The GSM module again provides real-time communication, sending alerts to appropriate personnel to prevent accidents or damage. This system helps monitor environmental hazards and structural risks, ensuring that workers are alerted in advance to dangerous situations.

3.3.1 Circuit Diagram of Manhole Monitoring System

The Figure 3.4 illustrates the circuit diagram of a Manhole Monitoring System, designed to monitor various environmental parameters within a manhole. The system integrates multiple sensors and components to provide real-time data and ensure the safety of workers operating in hazardous environments. The primary sensors in the circuit include an oxygen sensor, which detects the level of oxygen inside the manhole, and an MQ4 sensor, which measures the concentration of methane gas, a potentially dangerous gas in confined spaces.

In the Figure 3.4, a DHT11 sensor is used to monitor the temperature and humidity levels within the manhole.

These sensors provide vital information about the environmental conditions inside the manhole, helping to assess whether it is safe for workers to enter or remain in the area. The system also includes communication and processing units such as the SIM800L GSM module, which transmits data to remote locations via cellular networks, and the ESP32 microcontroller, which acts as the central processing unit, handling sensor data and communication. Two motors are also present, possibly for automated ventilation or other mechanical actions based on the sensor readings. This comprehensive setup ensures that gas levels, temperature, humidity, and other critical environmental factors are continuously monitored and communicated to external systems, providing a crucial safety mechanism for manhole operations.

The ESP32 microcontroller plays a pivotal role in the system's functionality. Figure 3.4 gives a low-power, high-performance controller with built-in Wi-Fi and Bluetooth capabilities, the ESP32 handles the real-time data collection from the sensors and processes it for analysis. Its versatile communication interfaces allow seamless integration with both the local system and remote servers. This enables the continuous relay of critical information about manhole conditions to a central monitoring station or a mobile device, ensuring rapid response to hazardous situations. The SIM800L module further enhances the system's communication capabilities by providing GSM connectivity, allowing the system to operate in areas with limited Wi-Fi or network coverage.

This feature is essential in underground or remote locations, ensuring that alerts or notifications about dangerous gas concentrations, oxygen depletion, or extreme temperature and humidity levels are sent immediately to workers or supervisors. This helps prevent accidents and ensures timely intervention if conditions inside the manhole become unsafe.

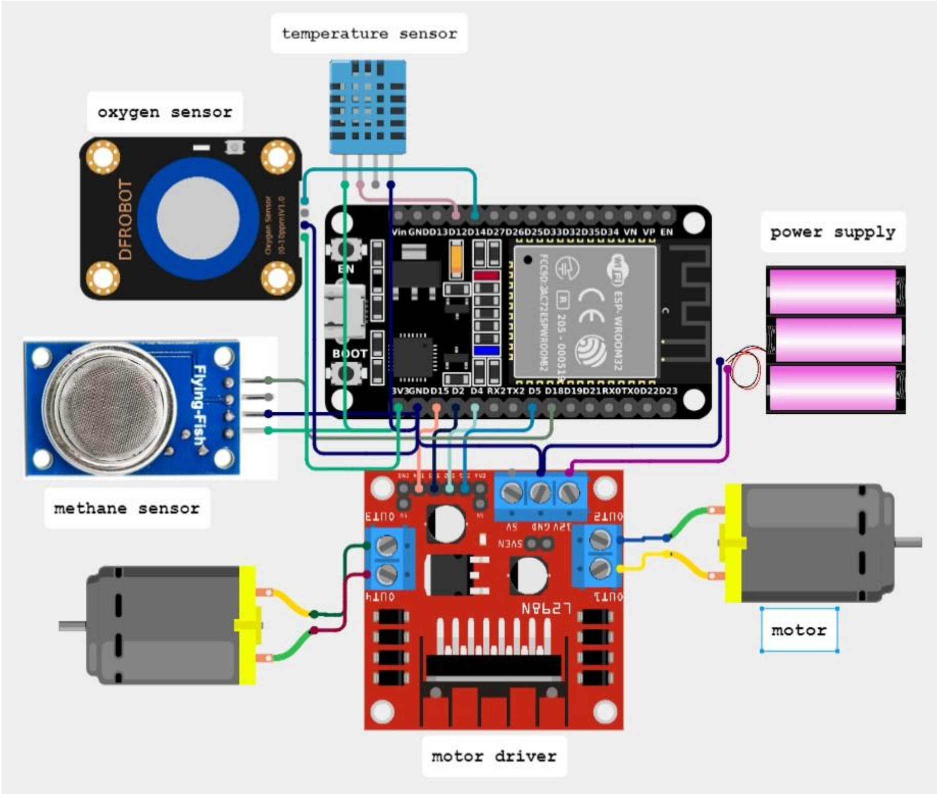

Figure 3.4: Manhole Monitoring System Circuit Diagram

The presence of two motors in the circuit diagram could be for mechanical operations, such as automated lid opening/closing or ventilation fan control, further enhancing the safety of the environment. By automating these functions, the system minimizes the need for workers to manually perform potentially dangerous tasks, reducing their exposure to harmful conditions. This comprehensive design ensures that the system can effectively monitor, communicate, and act upon the various hazards within a manhole, providing an essential safety net for underground workers.

3.3.2 Circuit Diagram of Manhole Maintainance Worker Monitoring System

The Figure 3.5 represents the Manhole Maintenance Worker Monitoring System. This system is designed to track the health and safety of workers involved in manhole inspections and maintenance. The core components in this system include the ESP32 microcontroller, AD8232 sensor, DHT11 sensor, SPO2 sensor, and the SIM800L GSM module. The ESP32 microcontroller serves as the central processing unit for this system. It collects and processes data from various health-monitoring sensors attached to the worker.

The AD8232 sensor is responsible for measuring the worker's heart rate and ECG signals, ensuring that the cardiovascular condition of the worker is monitored in real time.

Meanwhile, the SPO2 sensor measures the blood oxygen saturation levels, a key indicator of respiratory health, especially important when working in confined spaces where oxygen levels could

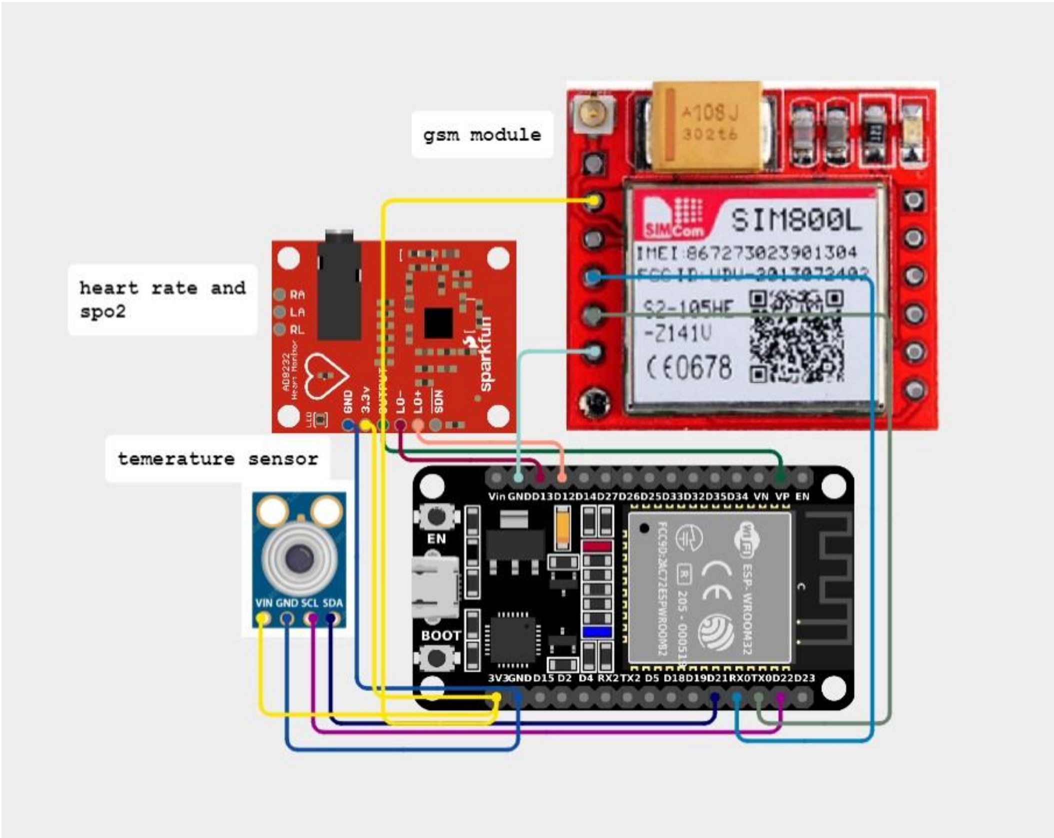

Figure 3.5: Manhole Maintainance Worker Monitoring System Circuit Diagram

drop. The DHT11 sensor, integrated into the worker's suit or environment, monitors the ambient temperature and humidity levels. High temperatures or humidity levels can lead to dehydration or heat-related illnesses, making this data crucial for maintaining the worker's safety in challenging environments.

The SIM800L GSM module in Figure 3.5 enables communication by transmitting the collected health and environmental data to a remote server or a supervisor's device. Figure 3.5 enables real-time monitoring system ensures that if any critical parameters, such as abnormal heart rates, low blood oxygen levels, or extreme environmental conditions, are detected, immediate alerts are sent. This allows for prompt intervention, safeguarding the health and

well-being of the workers during manhole maintenance tasks.

The Manhole Maintenance Worker Monitoring System enhances worker safety by providing continuous surveillance of vital signs and environmental factors, significantly reducing the risks involved in working within hazardous conditions. The system's ESP32 microcontroller plays a crucial role in ensuring seamless integration and communication between sensors and the external network. The collected data is analyzed in real-time, offering precise monitoring of the worker's health status. The AD8232 sensor tracks the worker's electrocardiogram (ECG), providing detailed information about heart activity, which is essential for detecting signs of fatigue, stress, or potential cardiac events during strenuous work in confined spaces.

Along with the SPO2 sensor, which measures oxygen saturation levels in the blood, the system ensures that respiratory issues such as hypoxia, common in low-oxygen environments like manholes, are quickly identified.

The DHT11 sensor contributes by monitoring the surrounding environmental conditions such as temperature and humidity. These factors are critical because excessive heat or humidity can lead to heatstroke, dehydration, or other heat-related health issues, especially when the worker is wearing heavy protective gear. The SIM800L module not only enables real-time data transmission to a remote server but also allows for alerts to be sent directly to the health and safety supervisors or emergency responders. This feature ensures that any anomalies in the worker's vital signs or environmental conditions trigger an immediate response, minimizing the time to act during emergencies. The system is crucial for environments where timely medical intervention could mean the difference between safety and life-threatening situations.

3.3.3 Circuit Diagram of Manhole Lid Monitoring System

The Figure 3.6 illustrates the Manhole Lid Monitoring System, which focuses on ensuring the security and integrity of manhole covers. This system is crucial for preventing unauthorized access, theft, or displacement of manhole lids, which could lead to public safety hazards or operational disruptions. The system is built around an ESP32 microcontroller that acts as the central controller, gathering and processing data from the various sensors and ensuring real-time communication through the GSM module SIM800L. The Tilt Sensor S205WD is the key component that detects any movement or displacement of the manhole cover. If the lid is opened or tilted beyond a certain angle, the system immediately registers the change. This sensor ensures that any unauthorized access or accidental opening of the manhole cover is detected, helping to prevent accidents or malicious activity. Additionally, the Level Sensor monitors the presence of water or liquid levels inside the manhole. This is essential for detecting flooding or overflow situations, which can pose significant risks to infrastructure and public safety. By keeping track of water levels, the system can provide timely alerts in case of water ingress, enabling quick action to avoid potential damage or hazards. The SIM800L GSM module allows the system to send alerts or notifications via SMS or data services to a central monitoring station or designated personnel.

This ensures that any abnormal conditions, such as lid movement or rising water levels, are communicated in real-time, enabling swift intervention and maintenance responses. The overall system offers a robust solution for safeguarding manhole covers and ensuring operational safety in urban infrastructure. The Manhole Lid Monitoring System serves as a critical safeguard for urban infrastructure management. In many cases, manhole lids can be tampered with or removed without authorization, leading to safety risks such as accidents, injuries, or unauthorized entry into the sewer systems. This system effectively mitigates these

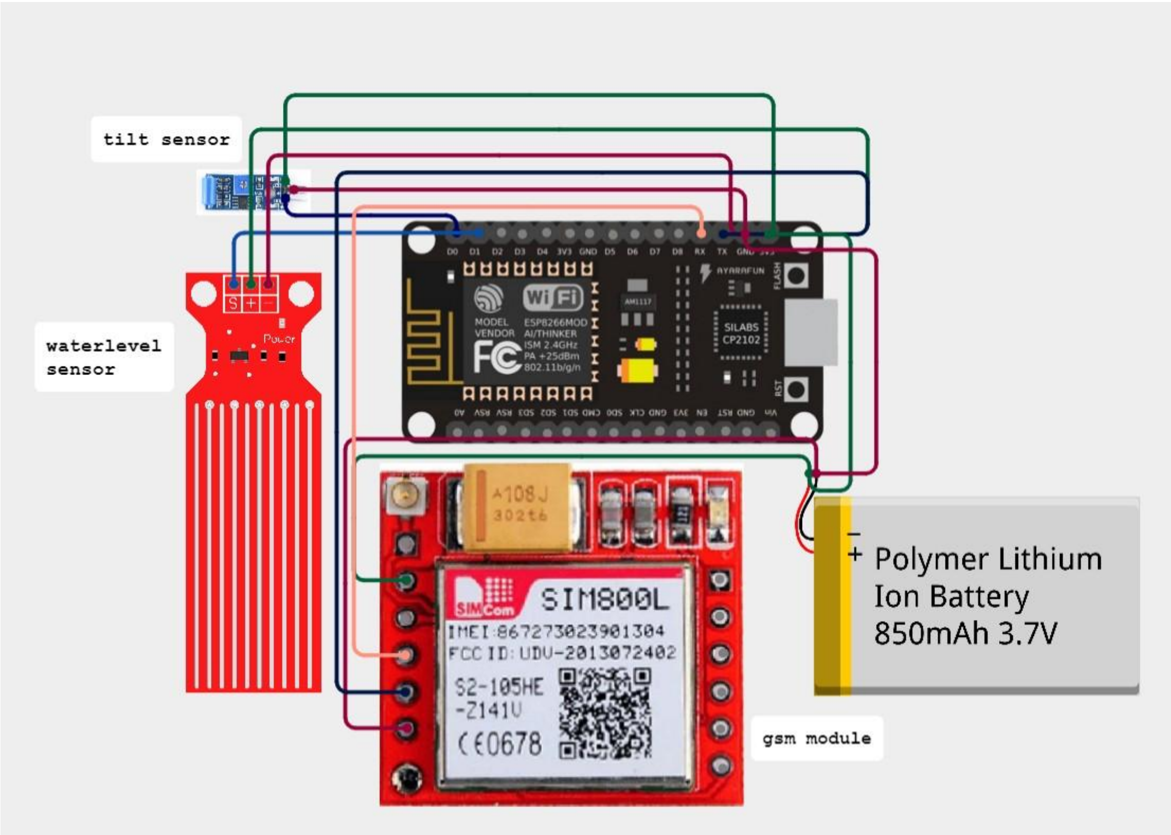

Figure 3.6: Manhole Lid Monitoring System Circuit Diagram

risks by integrating a combination of sensors and wireless communication technology to monitor the lid's condition continuously.

At the core of this system is the ESP32 microcontroller, a powerful and energy-efficient processor that manages all sensor inputs and ensures seamless connectivity. The Tilt Sensor is vital for detecting the slightest deviation in the orientation of the manhole lid, triggering an alert if the lid is moved. This helps in both detecting accidental displacement and preventing theft or vandalism, a common issue in many urban areas. Complementing this is the Water Level Sensor, which is particularly crucial in monitoring potential flooding. Manholes are often vulnerable to heavy rainfall or pipe bursts, and rising water levels inside can create significant infrastructure damage and pose a threat to nearby environments.

With real-time monitoring, the system in Figure 3.6 can alert relevant authorities before the situation escalates, reducing the risk of accidents, equipment damage, or environmental contamination. The inclusion of the SIM800L GSM module provides long-range communication capabilities, allowing the system to transmit critical data to remote servers or directly notify personnel through SMS or network data. This real-time alerting capability ensures that necessary actions, such as deploying maintenance crews or initiating safety protocols, can be taken promptly without the need for constant on-site monitoring. The system is powered by
a Polymer Lithium-Ion Battery, ensuring reliability and continued operation, even in cases where external power sources are disrupted.

3.4 Conclusion

An integrated IoT-based approach to monitoring both the health of manhole maintenance workers and environmental conditions within the manhole. The chosen methodology successfully
combines sensor-based data acquisition, real-time data processing, and wireless communication to ensure a comprehensive safety monitoring solution. By utilizing ESP8266 microcontrollers along with various health and environmental sensors such as PPG, MQ4 gas sensor, DHT11, and tilt sensors the system provides continuous and accurate measurements of both the
worker's physiological status and potential hazardous gases within the manhole.

The inclusion of the SIM800L GSM module for communication has proven effective for real-time alerts to authorities, ensuring that any critical changes in health metrics or environmental parameters are promptly reported. This methodology addresses the specific requirements for confined-space monitoring by integrating multi-sensor data and enabling remote alerts, providing a proactive solution to enhance safety and response times. The integration of the SIM800L GSM module for SMS notifications proved effective for real-time alerts, particularly in locations where Wi-Fi might be unavailable or unreliable. Cloud integration with the Blynk platform allowed for data visualization and trend tracking, ensuring that data was not only available for immediate monitoring but also for longer-term analysis. This dual approach supports both real-time intervention and ongoing safety management, demonstrating the versatility of IoT solutions for confined space monitoring.

During testing, the system effectively communicated sensor data to the cloud and triggered alerts when conditions exceeded set thresholds, validating the setup's reliability and confirming the threshold values for each parameter. The methodology also included rigorous testing of the SMS alert system, which showed consistent performance in delivering timely notifications. This approach ensured that each component met its intended function and was resilient under different operating conditions.

The provides a solid foundation for developing a robust and scalable IoT monitoring system, combining real-time data collection, reliable communication, and accessible cloud-based monitoring. The structured approach facilitated a streamlined implementation, highlighting the potential for further enhancements, such as expanding sensor types or integrating machine learning for predictive analytics, to advance confined space safety solutions.

CHAPTER 4

SYSTEMDESIGN

4.1 Introduction

Creating an IoT-based monitoring and alerting solution tailored for confined space safety. It integrates sensors, GSM communication modules, cloud processing, and automated alert mechanisms to monitor environmental parameters, such as temperature, toxic gas levels, and air quality, and alert personnel in case of hazards. Key components include sensing modules that gather real-time data on critical safety metrics, a GSM module to transmit this data to a cloud platform for remote access, and emergency alert functions that send SMS, email, or mobile app notifications when parameters exceed safe limits. The cloud platform acts as a central hub, storing and analyzing data to support both real-time monitoring and long-term trend analysis, which can highlight potential risks. A user-friendly dashboard displays real-time data, system status, and allows for custom configurations, such as setting threshold levels and managing alert recipients. A scalable, reliable solution to confined space monitoring, enhancing safety by ensuring immediate alerts, easy data access, and proactive monitoring for both current and future safety needs.

4.2 Hardware Used

4.2.1 ESP8266 Microcontroller

The ESP8266 is a popular, low-cost Wi-Fi microchip developed by Espressif Systems, widely used for Internet of Things applications. Known for its affordability and ease of use, the ESP8266 integrates a complete TCP/IP stack, enabling seamless network connectivity in a compact form factor. At its core, it features a 32-bit Tensilica L106 microcontroller operating at up to 80 MHz (with the ability to overclock to 160 MHz), providing sufficient processing power for a range of IoT applications.

This microcontroller is equipped with 64 KB of instruction RAM and 96 KB of data RAM, allowing it to perform various tasks independently. The ESP8266 shown in Figure 4.1 supports client and server Wi-Fi modes, making it versatile for use in smart home devices, security systems, wearable health monitors, and industrial automation. Equipped with an integrated stack, the ESP8266 handles basic network protocols, which facilitates web requests, socket connections, and data transmission without additional hardware. The module is available in several forms, such as the compact ESP-01, which has a limited number of pins, and the more versatile ESP-12, which offers greater functionality and antenna reception. Additionally, the ESP8266-based NodeMCU development board includes USB-to-serial converters and voltage

Figure 4.1: ESP8266 Microcontroller

regulators, making it even easier to program and use. Overall, the ESP8266 is widely used in applications ranging from home automation and weather stations to security and industrial monitoring systems. Despite newer alternatives like the ESP32, the ESP8266 remains a favored choice due to its balance of performance, affordability, and community support, making it ideal for projects where Wi-Fi connectivity and low power consumption are crucial.

One of the ESP8266's standout features is its support for low-power operation, with sleep modes that allow it to conserve energy, an essential capability for battery-powered IoT devices. In deep sleep mode, the chip can draw as little as 20 µA, which significantly extends battery life in remote applications. With up to 17 pins, the ESP8266 can interface with external sensors and devices, such as LEDs, relays, and buttons, although the specific number of pins available depends on the module variant. It also supports standard communication protocols
which enables connectivity with other microcontrollers, sensors, and actuators.

The ESP8266 can be programmed using multiple platforms like the Arduino IDE, MicroPython, and Espressif's own ESP, making it highly accessible to both beginners and advanced developers. Wi-Fi Capabilities: The ESP8266 is primarily known for its built-in Wi-Fi capabilities, supporting both client and server modes. It allows devices to connect to a

Wi-Fi network or create their own access point, making it highly versatile for IoT applications that require remote data transfer.

1. Microcontroller Unit: The ESP8266 contains a 32-bit Tensilica L106 microcontroller,which operates at a clock speed of up to 80 MHz (and can be overclocked to 160 MHz). It has 64 KB of instruction RAM and 96 KB of data RAM, allowing it to run a range of applications directly on the chip.

2. Power Efficiency: The ESP8266 is designed to be energy-efficient, supporting sleepmodes such as deep sleep and light sleep, which help reduce power consumption. In deep sleep mode, the chip can draw as little as 20 µA, making it ideal for battery-powered IoT applications.

3. Memory and Storage: The module typically has 4 MB of flash memory, though thiscan vary by module version. This storage is used for both code and data storage, and some modules allow expansion through an external flash chip.

4. GPIO Pins: The ESP8266 has up to 17 GPIO (General Purpose Input/Output) pins,which allow it to interface with other sensors, actuators, and peripherals like LEDs, relays, and switches. Certain modules, like the NodeMCU, expose more of these pins for user access.

5. Communication Protocols: The ESP8266 supports standard communication protocolsincluding UART, SPI, and I2C, which make it compatible with a wide range of external sensors and modules. Additionally, the ESP8266 can handle HTTP, MQTT, and WebSocket protocols, making it suitable for both web and networked applications.

6. Integrated TCP/IP Stack: With a built-in TCP/IP stack, the ESP8266 can handle basicnetwork protocols without additional support, enabling seamless HTTP requests, socket connections, and data transmission. This makes it perfect for IoT applications that require network communication with minimal coding.

7. Modularity and Variants: The ESP8266 comes in multiple module forms, such asthe ESP-01, ESP-07, and ESP-12. Modules like ESP-01 are more compact and offer fewer GPIO pins, while ESP-12 is more versatile, providing more GPIO access and better antenna reception.

8. Programming Support: The ESP8266 can be programmed with the Arduino IDE,MicroPython, and Espressif's own ESP-IDF development framework. This flexibility makes it accessible to both beginners and advanced users.

4.2.2 AD8232 ECG Sensor

The AD8232 is an analog front-end module created by Analog Devices, specifically engineered for biopotential measurements such as electrocardiography and heart rate monitoring. Compact and efficient, the Figure 4.2 is a AD8232 that amplifies and filters weak cardiac signals, which are susceptible to noise, to provide clean data on the heart's electrical activity. This makes it an ideal solution for wearable and portable health monitoring systems, especially in applications that require low power consumption, such as fitness trackers and medical IoT devices.

One of the AD8232's main strengths lies in its low power consumption, drawing only around 170 µA, which allows it to function effectively in battery-operated environments. This characteristic makes it well-suited for mobile health applications, where extended battery life

Figure 4.2: AD8232 ECG Sensor

is critical. Additionally, the AD8232 integrates essential signal conditioning, which includes amplification and filtering of ECG signals.

Its built-in instrumentation amplifier rejects high-frequency noise and interference from muscle signals or other sources, allowing for accurate signal acquisition even in noisy environments. Designed for single-lead ECG monitoring, the AD8232 captures the heart's electrical signals using three electrodes that connect to the left arm, right arm, and right leg. This setup provides basic heart rate data while maintaining a simple design for easy integration.

Developers can adjust the amplification level to isolate the ECG waveform, allowing it to be processed by a microcontroller. The AD8232's analog output is compatible with popular microcontrollers like Arduino, ESP32, and ESP8266, making it versatile for DIY projects and IoT health applications alike. The AD8232 is commonly found in applications that require compact, continuous cardiac monitoring, including wearable health devices like smartwatches and chest straps.

It is also used in telemedicine and fitness technology, where users or healthcare providers need real-time heart rate monitoring. However, as a single-lead ECG device, the AD8232 has limitations in providing comprehensive cardiac data compared to multi-lead systems. Additionally, since the data can be affected by motion artifacts, it is less ideal in environments with high movement.

In summary, the AD8232 module offers a practical and affordable solution for ECG monitoring in a wide range of personal and professional health applications. Its low power,

Figure 4.3: DSB18B20 Temperature Sensor

simple integration, and effective signal conditioning make it a valuable component in IoT health and fitness technology, enabling users to track their cardiovascular health conveniently.

4.2.3 DS18B20 Temperature Sensor

The DS18B20 is a digital temperature sensor widely used in a range of applications, from environmental monitoring to industrial systems, due to its accuracy, reliability, and ease of use. Manufactured by Maxim Integrated, this sensor can operate in temperatures between -55°C and +125°C, with an accuracy of ±0.5°C for most of this range. The outputs of the sensor shown in Figure 4.3 gives temperature data in a digital format, making it highly compatible with microcontrollers and microprocessors used in IoT and embedded systems. One of the standout features of the DS18B20 is its 1-Wire communication protocol, which allows data transfer and power to be supplied over a single data line. This protocol simplifies wiring, especially in applications with multiple sensors, as multiple DS18B20 sensors can be connected to the same data line while still being uniquely addressable. Each DS18B20 has a unique 64-bit serial code, allowing precise identification in a multi-sensor network, a valuable feature in systems requiring multiple temperature measurements, such as weather stations or HVAC systems.

The sensor operates in both parasitic power mode and external power mode. In parasitic mode, the sensor draws power directly from the data line, eliminating the need for an additional power source, which is particularly useful in low-power or compact applications. When connected to an external power source, the DS18B20 can achieve faster conversion times, an advantage in applications where real-time data is essential.

The DS18B20 also features programmable resolution between 9 and 12 bits, allowing developers to balance between precision and power consumption based on their specific needs. At its highest 12-bit resolution, it can measure temperatures in increments as small as 0.0625°C. This flexibility makes it suitable for applications that demand both high precision and efficiency, such as laboratory equipment, cold chain monitoring, and household appliances.The DS18B20 is a versatile and robust temperature sensor ideal for both hobbyist and professional applications.

Its digital output, unique addressability, and simple wiring make it easy to integrate into complex systems, while its wide temperature range and accuracy allow for reliable temperature measurement in a variety of environments. Whether used in home automation, industrial monitoring, or scientific applications, the DS18B20 remains a popular and reliable choice for temperature sensing.

4.2.4 SIM 800L GSM Module

The SIM800L GSM module is a compact, cost-effective solution that enables devices to communicate over a mobile network, allowing for voice calls, SMS messaging, and internet connectivity through GPRS. Manufactured by SIMCom, this module is especially popular in IoT applications where remote communication is essential, such as in smart meters, vehicle tracking, and remote environmental monitoring.

One of the main advantages of the SIM800L shown in Figure 4.4 is its GSM Quad-Band support, operating on frequencies 850, 900, 1800, and 1900 MHz. This ensures that it can work globally, providing connectivity in various locations worldwide where GSM networks are available. With a small form factor, it is easy to integrate into compact devices or systems where space is limited.

The module supports SMS and voice calls, making it useful for basic communication needs. The SMS functionality is especially advantageous for applications where data usage is minimal, and basic messaging will suffice, like alert notifications in security systems or remote status updates. In addition, the module can connect to the internet using GPRS (General Packet Radio Service) with speeds up to 85.6 kbps, allowing for low-bandwidth data transmission for applications such as telemetry and remote data logging.

Powering the SIM800L requires a stable 3.4V to 4.4V power supply. Due to its peak current consumption during network operations, it often requires a high-current power source, especially during data transmissions or call initiation. Because of this, careful power management and a capacitor to stabilize power are recommended to avoid performance issues.

The SIM800L connects to microcontrollers through Universal Asynchronous Receiver/Transmitter, making it straightforward to interface with devices like the Arduino, ESP32, and Raspberry Pi. Commanding the module is also simple via AT commands, which control everything

Figure 4.4: SIM 800L GSM Module

from setting up an SMS to connecting to the internet. This standardized communication makes it accessible and flexible for various applications, including vehicle tracking, remote environmental monitoring, home automation, and IoT projects that need cellular connectivity.

4.2.5 MQ4 Methane Sensor

The MQ4 methane sensor is specifically designed for detecting methane and other combustible gases such as and natural gas, making it an ideal component in gas leak detection systems and safety applications. This sensor can detect gas concentrations ranging from 200 to 10,000 ppm, which suits it for environments requiring both low- and high-level gas detection. With its versatility, the MQ4 finds applications in industrial safety, home safety, and environmental monitoring, where accurate methane measurement is essential.

Inside the sensor, visualising the Figure 4.5 a heating element works alongside a tin dioxide sensing layer to enhance sensitivity and provide stable measurements. This heating element ensures the sensor maintains a constant temperature, which improves accuracy and performance. However, the sensor does require a warm-up period of a few minutes to reach stable readings after being powered on. Its dual output capabilities—both analog and digitalallow it to offer real-time concentration levels while also triggering alarms when a preset gas concentration threshold is reached. This makes it versatile and compatible with microcontrollers, including Arduino, ESP8266, and ESP32, enhancing its ease of integration into various systems.

The MQ4 sensor is commonly used in gas leak detection for residential and industrial applications. In addition to detecting methane leaks, it is valuable in kitchens, factories, and laboratories, where it can activate alarms or initiate safety protocols if dangerous gas levels

Figure 4.5: MQ4 Methane Sensor

are detected. Moreover, due to its affordability, the MQ4 is widely adopted in projects and research, especially where continuous or remote monitoring of methane is needed.

While the MQ4 sensor is effective, it has certain limitations. The sensor is sensitive to changes in humidity and temperature, which may impact its accuracy if it operates in fluctuating environmental conditions. Additionally, it draws a significant amount of power due to the internal heating element, which may not be ideal for battery-operated applications. Despite these challenges, the MQ4 remains a valuable sensor due to its performance, cost-effectiveness, and compatibility with various microcontroller systems, making it widely used in gas sensing applications.

4.2.6 DHT11 Temperature Sensor

The DHT11 is a basic, affordable digital sensor used to measure both temperature and humidity, making it ideal for a range of indoor environmental monitoring applications. Its simplicity, small size, and ease of use have made it popular in home automation, greenhouse monitoring, and weather station projects. The DHT11 provides data in a digital format, which is convenient for microcontrollers, eliminating the need for additional components to interpret the data. The temperature measurement range of the DHT11 is 0–50°C with an accuracy of ±2°C, while its humidity range spans 20–90 with an accuracy of ±5.

While not as precise or responsive as other sensors, the DHT11 performs reliably for general-purpose applications where extreme precision is not required. The sensor can sample data once every second, which is adequate for monitoring environments with gradual changes. Connecting the DHT11 to microcontrollers like Arduino or ESP32 is straightforward, thanks to its digital signal output. The DHT11 operates on low power, typically running at 3.3–5V,

Figure 4.6: DHT11 Temperature Sensor

which makes it suitable for battery-powered devices as well. It uses a single data line to transmit temperature and humidity readings to the microcontroller, which reduces wiring complexity.

The DHT11 does have certain limitations. This temperature sensor in Figure 4.6 does not suitable for applications that require high-precision readings or fast sampling rates. Additionally, it is sensitive to extreme conditions, and performance can degrade if exposed to high humidity or temperature levels outside its specified range.The DHT11 is a practical and cost-effective sensor for basic temperature and humidity monitoring. Its ease of use and low power consumption make it an excellent choice for hobbyist projects, home automation systems, and general-purpose environmental sensing where basic accuracy is sufficient.

4.2.7 Tilt Sensor

A tilt sensor is a device designed to detect the orientation or tilt of an object. It operates by sensing changes in the angle or position of an object relative to gravity, making it valuable for a variety of applications, including equipment monitoring, automotive systems, and consumer electronics. Tilt sensors are often used in systems where monitoring the movement or angle of an object is necessary, such as in security alarms, gaming controllers, and industrial machinery.

Tilt sensors typically consist of an internal mechanism that shifts in response to changes in position. Many tilt sensors use a rolling ball, mercury, or conductive liquid inside a chamber.
When the sensor's angle changes, the internal material moves, completing or breaking a

Figure 4.7: Tilt Sensor

circuit. This shift in the sensor's internal state produces an electrical signal that indicates the tilt angle. In digital tilt sensors, the output is typically binary, signaling whether the sensor is tilted or not, making it easy to interface with microcontrollers. More advanced tilt sensors, like MEMS (Micro-Electro-Mechanical Systems) accelerometers, can provide precise angle measurements and detect multi-axis orientation.

Tilt sensor shown in Figure 4.7 are commonly applied in safety and security systems to monitor movement or tampering. For instance, they are used in vehicle anti-theft systems to detect unauthorized lifting or towing. In industrial applications, tilt sensors help monitor the alignment of machinery, improving operational safety and efficiency. They are also found in consumer electronics, such as smartphones and gaming controllers, to enable motion-based interaction. In summary, tilt sensors are essential for applications that require reliable detection of an object's angle or movement. Their simplicity, durability, and responsiveness make them ideal for various safety, monitoring, and consumer applications, providing accurate orientation information and enhancing device interactivity.

4.2.8 Water Level Sensor

A water level sensor is a device that detects and measures the level of water or other liquids in a container, tank, or natural body of water. These sensors play a crucial role in applications that require liquid level monitoring, including water tanks, reservoirs, wastewater management, and environmental monitoring. By providing real-time water level data, water level sensors are instrumental in industries like agriculture, manufacturing, and smart home automation. They are also widely used in IoT-based systems where data on liquid levels can inform automated processes or alert users about critical conditions.

Figure 4.8: Water Level Sensor

Water level sensor in Figure 4.8 operate in different ways depending on the technology used. Float sensors, for instance, use a simple floating mechanism that rises and falls with the liquid level. This motion activates a switch, which can signal if the liquid is above or below a certain threshold. These sensors are highly reliable and popular for their simplicity, often found in home water tanks and some industrial applications. In contrast, capacitive sensors detect changes in capacitance caused by the proximity of water. This type of sensor is non-contact, meaning it does not touch the liquid, making it ideal for industries like food and pharmaceuticals where contamination is a concern. Other types include conductive sensors,
which use electrodes to detect the liquid level based on its conductivity.

When water contacts a certain electrode, it completes an electrical circuit, allowing the sensor to detect the level. Conductive sensors are cost-effective and often used in tanks and reservoirs where high precision is not required. Lastly, ultrasonic sensors measure the water level by emitting ultrasonic waves and calculating the time it takes for the waves to bounce back from the water's surface. These non-contact sensors offer high precision and are frequently employed in applications requiring accurate, continuous level measurements, such as in environmental monitoring or large water storage facilities.

The output from a water level sensor can either be digital or analog. Digital sensors often provide a high or low signal, indicating if water is present or absent at a specific level. In contrast, analog sensors offer a variable output corresponding to the water level, which allows for more detailed monitoring. This versatility in output types enables water level sensors to be used in a wide variety of systems, from simple alerting mechanisms to complex monitoring systems requiring nuanced data.

Figure 4.9: O2-A2 Oxygen Sensor

Water level sensors are versatile and essential devices for monitoring liquid levels. With options tailored to various needs—whether simple, cost-effective solutions or high-precision instruments—water level sensors support a range of applications from industrial water management to household automation, contributing to safety, efficiency, and resource conservation across numerous fields.

4.2.9 O2-A2 Oxygen Sensor

The O2-A2 standard serves as a crucial guideline for the design, testing, and quality assurance of oxygen sensors utilized in automotive applications. These sensors are essential for vehicles equipped with internal combustion engines, where they monitor the oxygen concentration in the exhaust gases. The primary role of the oxygen sensor is to provide real-time feedback to the engine control unit, enabling it to adjust the air-fuel mixture.

This optimization is vital for achieving efficient combustion, which not only enhances fuel economy but also minimizes harmful emissions that contribute to air pollution. A significant component of the O2-A2 standard is its emphasis on comprehensive testing procedures. The standard specifies various performance criteria that oxygen sensors must meet to ensure reliability and accuracy under different conditions.

Sensors are subjected to rigorous testing across a range of temperatures and pressures to simulate real-world operating environments. These tests also evaluate the sensors' responses to changes in exhaust gas compositions, including variations caused by different fuel types and engine loads. By defining these testing protocols, the O2-A2 standard helps manufacturers produce sensors that consistently deliver accurate readings, which are critical for optimal engine performance.

The O2-A2 in Figure 4.9 gives a standard underscores the necessity of robust quality control measures throughout the manufacturing process. This aspect of the standard is designed to ensure that every oxygen sensor produced adheres to strict guidelines regarding materials, design, and durability. Sensors are often exposed to extreme conditions, such as high temperatures, vibrations, and corrosive substances within the exhaust system. Consequently, the standard calls for manufacturers to implement rigorous inspection and testing protocols to guarantee that each sensor can withstand these challenges over its operational lifespan.

By maintaining high-quality standards, manufacturers can reduce the likelihood of sensor failure, which can lead to poor engine performance and increased emissions. Furthermore, the O2-A2 standard addresses environmental considerations that may impact the functionality of oxygen sensors. The standard acknowledges that contaminants present in the exhaust gases can affect sensor accuracy and longevity. Therefore, it encourages the development of sensors that are resistant to fouling and degradation, ensuring reliable operation even in the presence of harmful substances. This focus on durability and resilience is increasingly important as automotive technologies evolve, particularly with the growing emphasis on reducing emissions and meeting stringent environmental regulations.

The O2-A2 standard is essential for manufacturers aiming to produce high-quality oxygen sensors that contribute to improved vehicle performance and efficiency. The standard not only aids in enhancing fuel economy and reducing emissions but also plays a significant role in supporting the automotive industry's shift toward more sustainable practices. By following established guidelines, manufacturers can ensure their products align with regulatory requirements and meet the demands of modern vehicle technologies, paving the way for innovations that benefit both consumers and the environment.

4.2.10 L298N Motor Driver

The L298N is a widely used dual H-bridge motor driver integrated circuit that enables the control of DC motors and stepper motors in various applications. This IC is particularly popular in robotics and automation projects due to its ability to control two motors independently. The L298N allows users to drive motors in both forward and reverse directions, making it a versatile choice for applications that require precise motor control. One of the key features of the L298N in Figure 4.10 is its dual H-bridge configuration, which facilitates the control of two DC motors or one stepper motor. Each motor can be controlled separately, allowing for complex movements and operations. The L298N operates within a voltage range of 5V to 35V and can handle a maximum current of up to 2A per channel. This capability makes it suitable for small to medium-sized motors, making it an excellent choice for hobbyists and professionals alike. The motor control inputs of the L298N are straightforward, with several pins dedicated to controlling the direction of the motors. The inputs allow users to set the direction of rotation for each motor. Additionally, the enable pins can be utilized

Figure 4.10: L298N Motor Driver

for speed control through Pulse Width Modulation. By varying the duty cycle of the signal, users can effectively control the speed of the motors, providing a flexible solution for various motor-driven tasks.

In terms of connectivity, the L298N is typically available in a convenient breakout board format, making it easy to integrate with microcontrollers like Arduino or Raspberry Pi. The wiring connections are user-friendly; the VCC pin connects to the motor power supply, while the GND pin is connected to the ground of both the power supply and the microcontroller. This simplicity in design allows for quick setup and experimentation, making the L298N an ideal choice for prototyping and educational purposes. Furthermore, the L298N includes built-in thermal protection features to prevent overheating during operation. This thermal shutdown capability enhances the reliability and longevity of the IC, allowing it to operate effectively even under demanding conditions. Overall, the L298N motor driver is a powerful and flexible tool that facilitates precise motor control, making it an essential component in numerous robotics and automation projects.

4.3 Software Used

4.3.1 Arduino IDE

The Arduino IDE is designed to simplify the process of programming Arduino microcontrollers, making it accessible to both beginners and experienced developers. The environment supports a simplified version of C/C++ programming language, which allows users to write code using straightforward syntax and functions tailored for Arduino boards. The IDE includes built-in libraries that provide a wide range of functionalities, from controlling motors to handling sensors, enabling users to quickly develop applications without needing in-depth programming knowledge. One of the standout features of the Arduino IDE is its user-friendly interface.

The IDE includes a text editor for writing code, a message area for displaying information about the code compilation and upload processes, and a console for displaying debug messages. Users can easily open and manage multiple sketches (Arduino programs) and access a library manager to include external libraries, enhancing the capabilities of their projects. The IDE also features a simple one-click process to compile the code and upload it to the connected Arduino board. The Arduino IDE supports a wide variety of Arduino boards and compatible microcontrollers, including popular models like the Arduino Uno, Mega, Nano, and Leonardo.

To work with a specific board, users can select the board type from the IDE's menu, ensuring that the code is correctly compiled for the hardware in use. The IDE also includes a serial monitor, which allows users to send and receive data between the computer and the Arduino board, making it invaluable for debugging and monitoring the performance of their projects.

The Arduino IDE is highly extensible, allowing users to add libraries and third-party tools to enhance their development experience. There is a rich ecosystem of libraries available, covering various applications like IoT, robotics, and home automation. Users can also create custom libraries to modularize their code and reuse components across different projects. The Arduino community is vast and active, offering extensive online resources, forums, and tutorials to support users in their development journey.

The Arduino IDE is available for multiple operating systems, including Windows, macOS, and Linux, making it accessible to a wide range of users. The software is open-source, allowing developers to contribute to its improvement and adaptation. This openness has fostered a strong community that continuously shares knowledge, resources, and projects, further enriching the Arduino ecosystem. Overall, the Arduino IDE is an essential tool for anyone looking to explore the world of microcontrollers and embedded programming, enabling users to bring their creative ideas to life.

4.3.2 Firebase Platform

Firebase provides a wide range of features designed to enhance the development process and improve user engagement. One of its core offerings is the Firebase Realtime Database,
which allows developers to store and synchronize data in real-time across all clients. This is particularly beneficial for applications requiring live data updates, such as chat applications or collaborative tools. The database uses a NoSQL data structure, enabling developers to store data in JSON format, making it easy to access and modify. Another key feature of Firebase is Cloud Firestore, a more flexible and scalable database solution that supports richer data models and advanced querying capabilities. Firestore allows for hierarchical data organization, enabling developers to structure their data in collections and documents.

This flexibility makes it suitable for a wide range of applications, from simple to complex data requirements. Both databases provide offline capabilities, allowing users to access and manipulate data even without an internet connection. Firebase also includes Firebase Authentication, which simplifies the process of user authentication. It supports various sign-in methods, including email/password, phone authentication, and social media logins (such as Google, Facebook, and Twitter). This robust authentication system allows developers to easily manage user accounts and secure access to their applications, enhancing user experience and security.

For app performance monitoring and analytics, Firebase offers Google Analytics for Firebase, which provides detailed insights into user engagement and behavior within the app. Developers can track user interactions, measure app performance, and analyze conversion rates to make informed decisions about improvements and marketing strategies. The platform also includes tools for A/B testing and remote configuration, enabling developers to optimize their applications based on real-time user feedback.

Firebase provides a variety of cloud services through Firebase Cloud Functions. This feature allows developers to run backend code in response to events triggered by Firebase features and HTTPS requests. By utilizing serverless architecture, developers can focus on writing code without managing server infrastructure, leading to increased scalability and reduced operational costs. Firebase is also equipped with tools for push notifications through Firebase Cloud Messaging, enabling developers to engage users with targeted notifications and updates.

This feature is crucial for maintaining user retention and enhancing app engagement, as it allows for timely communication of relevant information. Overall, Firebase is a powerful platform that simplifies the app development process by providing a comprehensive suite of tools and services. Its integration with Google Cloud services ensures scalability and reliability, making it a popular choice for startups, enterprises, and independent developers alike. By leveraging Firebase, developers can focus on building high-quality applications that meet user needs while taking advantage of advanced features and capabilities.

4.4 Conclusion

The IoT-based monitoring and alerting system offers a robust, scalable solution for enhancing safety in confined spaces. By integrating real-time environmental sensing, reliable GSM communication, cloud-based data processing, and automated alert mechanisms, the system ensures that potential hazards are quickly identified and addressed. This approach not only enables rapid response through instant notifications to personnel but also supports preventive safety measures through long-term data analysis and trend monitoring. The system's design prioritizes accessibility and usability, providing remote monitoring capabilities and a user-friendly dashboard for real-time insights and system management. This proactive and data-driven framework effectively reduces risks, promotes timely intervention, and creates safer working environments in high-risk, confined spaces. Ultimately, this project demonstrates how IoT and cloud integration can transform safety practices, offering a comprehensive solution that safeguards worker health and improves operational resilience.

CHAPTER 5

RESULTS&DISCUSSION

5.1 Introduction

This provides an overview of the outcomes from testing the manhole monitoring and worker health monitoring systems. The project aimed to create a reliable system that could continuously monitor a maintenance worker's health and detect hazardous gas levels inside a manhole, sending alerts to authorities in case of dangerous conditions. This section will summarize how effectively each subsystem gas detection, health monitoring, and manhole condition tracking performed during trials, giving an initial insight into the system's accuracy, responsiveness, and reliability. Results show the system's effectiveness in detecting critical conditions like hazardous gas buildup, abnormal worker vitals, and manhole cover tampering, each of which is promptly communicated through the SIM800L GSM module. The accuracy of sensors such as the MQ4 gas sensor, AD8232 ECG sensor, and DHT11 temperature and humidity sensor was assessed, along with the overall communication reliability between the ESP32 microcontroller and GSM module. By capturing these critical safety parameters, the system demonstrated its capability to alert relevant authorities when necessary, meeting the primary objectives of the project. This summary of results paves the way for detailed discussions on the system's efficiency, real-time response, and areas for further improvement.

5.2 Analysis of the Proposed System

5.2.1 Analysis of the PPG Signal

A photoplethysmogram signal is a type of biomedical signal that measures changes in blood volume in the body's tissues, using a light source and a photodetector. It's commonly used to monitor heart rate, blood oxygen saturation , and heart rate variability. PPG works by shining light typically infrared or red into the skin. Blood absorbs this light, and as blood volume in the capillaries changes with each heartbeat, the amount of light absorbed also changes. The PPG sensor detects these changes, creating a signal that reflects the heart's activity. The PPG waveform has peaks and valleys: each peak corresponds to a heartbeat, where blood flow increases and light absorption is highest, while each valley shows reduced blood flow between beats, where light absorption decreases. The PPG signal has two main components. The AC component reflects the pulsatile changes in blood volume with each heartbeat, creating the characteristic waveform with peaks and valleys. This component is used to determine heart rate. The DC component represents the baseline signal level, related to the average blood volume in the tissue, and remains relatively constant. PPG is widely used

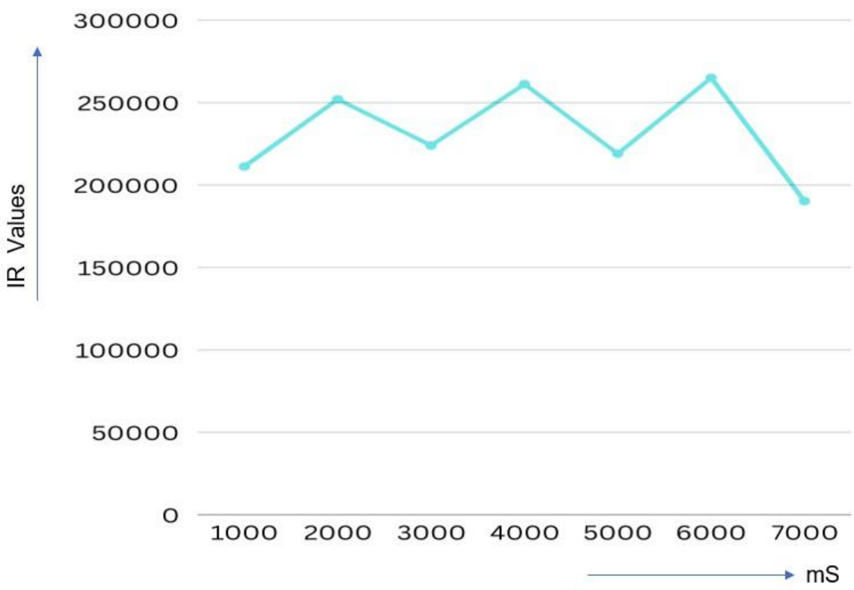

Figure 5.1: Analysis of the PPG Signal

to measure heart rate by counting the peaks in the signal, with each peak corresponding to one heartbeat. It can also estimate SpO2 by comparing the absorption of red and infrared light, as oxygenated and deoxygenated blood absorb light differently. Additionally, by analyzing the time intervals between consecutive peaks, PPG can measure heart rate variability, which provides insights into the autonomic nervous system and overall cardiovascular health.

This graph in Figure 5.1 is a representation of the photoplethysmogram (PPG) signal obtained from a MAX30102 sensor, showing changes in infrared (IR) light absorption over time. The x-axis represents time in milliseconds, ranging from 0 to 7000, with increments in thousands, meaning each major tick represents one second. The y-axis shows the IR values, which measure the intensity of infrared light absorbed by the blood. This absorption fluctuates with each heartbeat, providing a way to visualize the changes in blood flow and, consequently, the heart's activity over time.

The PPG signal, captured by the MAX30102 sensor, leverages the fact that blood absorbs infrared light differently depending on the blood volume in the capillaries, which varies with each pulse. As the heart pumps blood, there is a surge in blood volume that increases IR light absorption, creating peaks in the graph. Between beats, blood volume decreases, resulting in dips in absorption. This up-and-down wave pattern is characteristic of a PPG signal, where each peak represents a heartbeat and the intervals between peaks indicate the heart rate. In this particular graph, we see a consistent waveform that repeats approximately every 1000 milliseconds (or one second), suggesting a heart rate close to 60 beats per minute, which is typical for a resting heart rate. However, there is slight variability in the amplitude of the peaks, meaning the IR values fluctuate in magnitude with each pulse. This variability can be due to several factors, such as small changes in the sensor's position on the skin, slight motion artifacts, or natural variations in blood flow. Nonetheless, the overall periodicity remains stable, showing that the heart rate is relatively regular during this measurement period. Analyzing such a PPG signal can provide valuable insights into cardiovascular health.

By examining the height of the peaks (peak amplitude), we can infer changes in blood volume and potentially assess the condition of peripheral circulation. For instance, consistently low peaks might indicate poor blood perfusion. Additionally, the time interval between consecutive peaks (peak-to-peak interval) is directly related to heart rate variability (HRV), a critical metric in understanding autonomic nervous system activity. High HRV is generally associated with good cardiovascular fitness, while low HRV may indicate stress or other health concerns.

The MAX30102 sensor is widely used in wearable health devices because it can capture both IR and red light signals, which can help in determining blood oxygen saturation (SpO2) in addition to heart rate. SpO2 is calculated by comparing the absorption of IR and red light, as oxygenated and deoxygenated blood absorb these wavelengths differently. This graph, while only showing IR values, can still provide information on heart rate by counting the peaks over time and analyzing their rhythm and consistency. In a practical context, the data from this PPG signal could be fed into algorithms to detect irregular heart rhythms, such as arrhythmias, or to estimate blood oxygen saturation when combined with red light measurements.

Additionally, the amplitude and frequency of the peaks could be monitored over time to detect changes in cardiovascular health, which could be useful in managing chronic conditions or assessing the impact of physical activity. In summary, this graph of IR values over time represents the heart's pumping action, providing a non-invasive method to monitor heart rate and blood flow. The periodic peaks indicate consistent heartbeats, while the variability in peak height may reveal subtle physiological changes. This type of data is essential for modern wearable health technology, offering insights into heart rate and potentially other vital signs without the need for complex or invasive equipment.

5.3 Worker Monitoring System

This IoT-based health monitoring project is designed to provide continuous, real-time tracking of essential health metrics, specifically temperature, heart rate, and SpO2 (blood oxygen saturation). Using an ESP8266 microcontroller as its central processing unit, this system integrates three critical sensors: the DS18B20 for temperature, the AD8232 for ECG/heart rate, and the MAX30102 for SpO2 and pulse rate measurements. The system in Figure 5.2 incorporates both GSM and Blynk IoT connectivity, allowing for seamless data monitoring and communication. By providing both SMS alerts for immediate attention and a live dashboard on the Blynk app, this project becomes a comprehensive remote health monitoring solution ideal for various applications such as home healthcare, workplace safety, and even telemedicine setups.

The system's temperature monitoring capability is managed by the DS18B20 sensor, which measures ambient or body temperature with high precision. This digital sensor is calibrated for consistent readings and is ideal for continuous monitoring. In the project, a threshold of 40.0°C is set; if the sensor detects a temperature above this level, it triggers an SMS alert via the GSM module to a designated phone number, alerting users to the potential risk of fever or overheating.

The system sends temperature readings in real-time to the Blynk IoT app, where it's displayed on a virtual gauge. This remote access to live temperature data is highly valuable, enabling caregivers or family members to stay updated on the user's health status even from afar. For heart rate monitoring, the project uses the AD8232 ECG module. This sensor captures the heart's electrical signals and converts them into analog values that are read by the microcontroller. By processing these signals, the system calculates the beats per minute and detects irregularities in heart rhythm. It compares the BPM against a predefined safe range and, if the heart rate falls outside this range, an SMS alert is sent out. This functionality is particularly crucial for detecting arrhythmias or irregular heartbeats that might indicate a health issue. Alongside the SMS alert, real-time heart rate data is displayed on the Blynk app using virtual pin V1, offering immediate and continuous access to vital data that can be essential for individuals with known cardiac conditions.

The SpO2 monitoring feature is powered by the MAX30102 pulse oximeter, which calculates blood oxygen saturation by emitting red and infrared light into the skin and analyzing the absorption based on blood oxygen levels. This SpO2 value is critical for tracking respiratory and cardiovascular health, and it can reveal potential issues with oxygen intake or circulation. If the system detects an SpO2 level below the safe threshold of 95, an SMS alert is triggered, notifying users or caregivers that the blood oxygen saturation may be critically low. As with other metrics, the SpO2 data is displayed in real time on the Blynk app. This remote accessibility to oxygen saturation levels provides a convenient solution for consistent monitoring, especially for individuals at risk for respiratory issues or in recovery.

Communication and alerting are handled by the SIM800L GSM module, which sends SMS alerts whenever a reading exceeds or falls below the defined safety parameters. The use of GSM allows the system to function even in areas without Wi-Fi connectivity, ensuring critical alerts reach the designated phone number regardless of internet availability. For example, if the user's temperature exceeds 40°C or the SpO2 level drops below 95, an SMS alert is immediately sent, providing timely notifications of any health risks. Similarly, if the heart rate is detected outside the safe range, an alert is triggered to help facilitate swift intervention if needed. Additionally, Blynk IoT integration enables remote monitoring of all collected data through a user-friendly app interface.

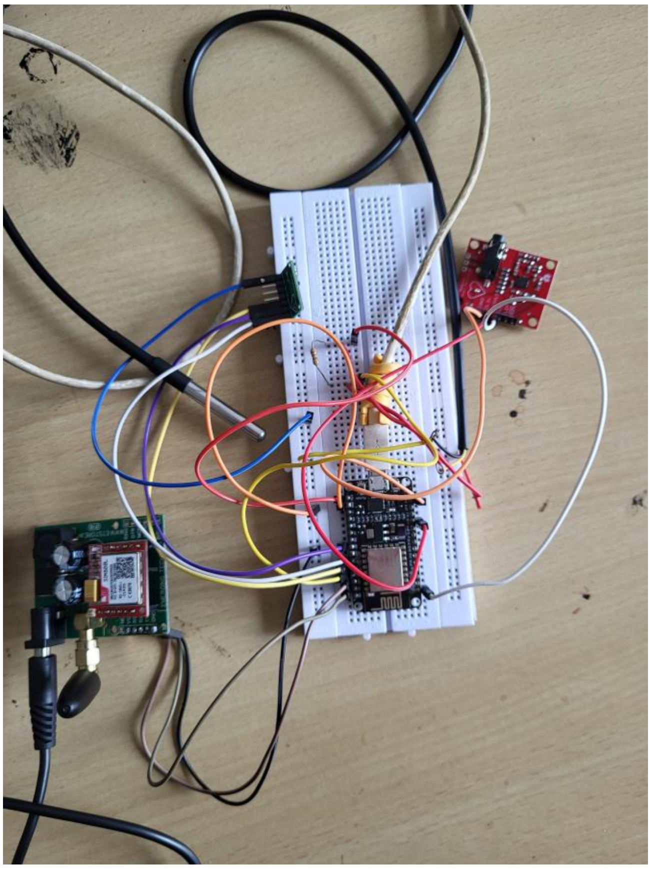

Figure 5.2: Worker Health State Monitoring System

The ESP8266 microcontroller maintains a continuous Wi-Fi connection to stream live data to the Blynk app, where the user can monitor temperature, heart rate, and SpO2 levels on virtual gauges. This remote access allows caregivers or family members to monitor vital signs from anywhere with an internet connection, offering an invaluable tool for overseeing a loved one's health without needing to be physically present. This feature is particularly useful in applications requiring close monitoring, such as elder care or health monitoring for individuals with chronic conditions.

The system is highly beneficial for health monitoring in multiple contexts. In home healthcare, it allows family members to keep track of an elderly or at-risk individual's vitals, providing peace of mind and ensuring timely intervention if readings are abnormal. For workplace safety, especially in high-risk environments, this project offers a way to ensure worker health by tracking key indicators that could signal overheating, dehydration, or respiratory strain. The alerting capabilities also make it suitable for use in outdoor or remote environments where internet access is limited. Additionally, this system is well-suited for integration into telemedicine frameworks, where healthcare providers can remotely access patient data and receive alerts for abnormal readings. This project demonstrates several key benefits, making it a well-rounded and valuable health monitoring solution. First, real-time health alerts through SMS ensure immediate action can be taken if a user's vitals enter critical ranges, providing an essential layer of safety. Remote monitoring via Blynk enhances accessibility, allowing caregivers and healthcare professionals to view health data and trends from anywhere as in the Figure 5.3. Furthermore, the cost-effective design makes it accessible, using affordable sensors and an open-source IoT platform that supports wide adoption.

The project is scalable and customizable, meaning additional sensors or personalized thresholds can be integrated to meet specific health needs or conditions. Overall, this IoT-based health monitoring system offers a reliable, accessible, and proactive solution for tracking temperature, heart rate, and SpO2. It combines real-time alerting, remote visualization, and easy expandability to provide a foundation for both personal and professional healthcare applications. Whether for home use, workplace safety, or telemedicine, this system provides a strong foundation for effective and continuous health monitoring.

5.3.1 GSM Alert Mechanism

The alert mechanism using the GSM module in this project is a critical feature that ensures users are promptly informed of any health risks or abnormal readings in real time, regardless of their access to Wi-Fi or the Blynk app. This functionality operates through the SIM800L GSM module, which is programmed to send SMS alerts directly to a designated phone number when specific thresholds for temperature, heart rate, or SpO2 levels are crossed. This offline alert capability provides peace of mind, especially in cases where the user or caregiver may not have immediate access to the Blynk app or internet connectivity, thus ensuring they are always informed about the user's health status.

The GSM alert mechanism is carefully designed to monitor the three primary health metrics individually. For temperature, if the DS18B20 sensor detects readings above the set threshold of 40°C, the GSM module sends an SMS stating "Alert: Temperature Warning!" This message is intended to prompt immediate action, potentially indicating a fever or other health risk that requires attention. For heart rate, the system continuously monitors the user's beats per minute (BPM) using the AD8232 ECG sensor, which detects any irregularities in the heart rate by checking if it falls below or exceeds the user-defined limits of 60 BPM and 120 BPM. Should an abnormal heart rate be detected, an SMS alert, "Alert: Heart Rate Warning!" is automatically triggered to notify the caregiver or family member of the need for a health check.

The SpO2 level is particularly important for detecting respiratory issues or oxygen deprivation, and the GSM module is programmed to send an "Alert: SpO2 Warning! Low Oxygen Level" SMS if the SpO2 reading drops below 96. This immediate notification enables caregivers to respond rapidly, potentially addressing critical health situations before they escalate. Each alert is configured to send the appropriate message with the GSM module's

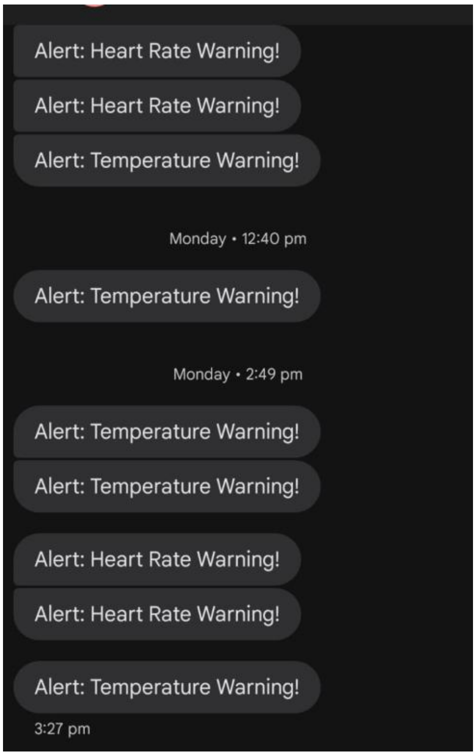

Figure 5.3: Alert Message Regarding Worker Health States

built-in AT command set, which initiates the SMS process by first setting the device to text mode and then issuing the command to send the SMS to the designated number.

The GSM alert system's design includes a debounce mechanism to prevent multiple alerts from being sent in quick succession due to sensor noise or brief fluctuations. For example, a delay is added between readings to ensure that alerts are only sent when sustained abnormal readings are detected, preventing unnecessary or duplicate alerts and ensuring that each notification reflects an accurate and critical change in the user's health.

5.3.2 Blynk Interface

The Blynk interface in this project provides a user-friendly, real-time platform for remote monitoring of key health metrics—temperature, heart rate, and SpO2—via an intuitive mobile app. Using virtual gauges, the interface presents each metric in a clear, readable format, with temperature displayed on Virtual PiN, heart rate and SpO2. This setup allows users to view the latest data for each metric in real-time, creating a visual snapshot of the user's current health status. The Blynk app's interface is designed to be responsive and customizable, enabling users or caregivers to monitor health metrics from anywhere with internet connectivity. The Blynk platform's cloud-based connectivity ensures that the ESP8266 device remains in sync with the app, automatically updating values as they are read from the sensors.

Real-time readings, the Blynk interface could be configured to track historical data, providing valuable trends that might help identify any concerning changes over time. The

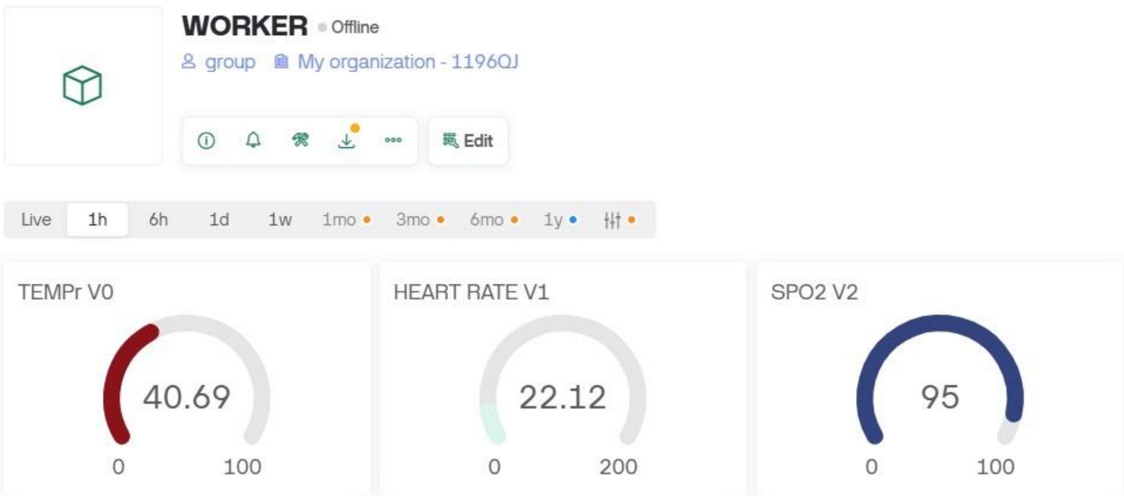

Figure 5.4: Blynk Interface

app's simple, reliable connection and remote accessibility make it highly suitable for caregivers monitoring patients in different locations. This functionality makes the Blynk interface a powerful component of the project, ensuring ease of use and peace of mind through seamless, on-demand access to essential health data. The figure 5.4 shows the Blynk Interface of the proposed system. The Blynk interface significantly enhances this project by bridging the gap between physical health sensors and remote, real-time monitoring. Once the ESP8266 microcontroller is connected to Wi-Fi, the Blynk app begins receiving live data updates, displaying them in a visually appealing format with adjustable gauge dials for temperature, heart rate, and SpO2. This visual feedback allows users to instantly grasp any fluctuations,

whether they're at home or monitoring a loved one from afar. The Blynk app's intuitive design offers an effortless experience, even for users with minimal technical background, making it accessible to a wide range of users, including healthcare professionals, caregivers, and individuals monitoring their own health. Additionally, the interface is flexible and can be customized further as needs evolve.

Custom alerts can be configured within Blynk, enabling app notifications for metric thresholds, such as a sudden rise in temperature or a drop in SpO2 levels. This ensures that even when users are not actively monitoring the app, they're still alerted to critical health changes. Users can add more gauges or modify existing virtual pins to monitor other health metrics, making Blynk an adaptable solution for various health needs.

The system's data synchronization across devices also allows multiple caregivers or family members to access the same live data, enhancing collaborative care and ensuring all relevant parties are updated in real time. Overall, the Blynk interface serves as an essential component, adding an extra layer of interactivity and accessibility to the project. Its design focuses on both clarity and functionality, allowing users to stay informed, set personalized alerts, and access health data anytime, anywhere.

5.4 Conclusion

The IoT-based monitoring system effectively tracks key health and environmental metrics, including temperature, heart rate, and SpO levels, within confined spaces. Through real-time data collection and processing, the system reliably alerts personnel when measurements exceed predetermined safety thresholds. SMS alerts and Blynk updates ensure that safety managers receive timely notifications, enabling rapid intervention when necessary. In testing, the temperature sensor reliably detected variations and triggered alerts at critical levels, while the heart rate and SpO sensors provided consistent readings that accurately reflected user conditions.

The integration with the SIM800L module proved effective for remote SMS notifications, enhancing responsiveness in emergency scenarios. The Blynk platform further facilitated remote monitoring by providing a clear, user-friendly interface for real-time data display.The system performed well in delivering real-time alerts and remote monitoring capabilities. These results validate the effectiveness of IoT and GSM technologies in enhancing confined space safety by providing reliable, continuous monitoring and swift notifications, demonstrating its potential for practical applications in various high-risk environments.

The implementation of this IoT-based monitoring system yielded positive results, confirming its capability to address critical safety challenges in confined spaces. The system's continuous monitoring and reliable alerting functions offer a proactive approach to safety management,

with each component contributing to its robust performance. During testing, the temperature sensor was highly responsive, detecting fluctuations with minimal latency and consistently triggering SMS alerts upon exceeding the safety threshold. This function is vital for preventing overheating risks in confined spaces.

The GSM module performed effectively for remote SMS based alerts, ensuring that notifications could reach personnel regardless of the availability of Wi-Fi. This capability is crucial in underground or remote locations, where cellular connectivity is the most reliable communication medium. Blynk integration allowed for real-time monitoring via a mobile-friendly interface, making it accessible to users on the move and providing a centralized view of all monitored parameters.

CHAPTER 6

CONCLUSION&FUTURESCOPE

6.1 Conclusion

The designed system for monitoring a manhole maintenance worker's health and the environment within manholes using IoT technologies provides a comprehensive solution for enhancing safety and efficiency. The integration of IoT sensors enables real-time monitoring of the worker's vital signs, such as heart rate, body temperature, and oxygen levels, allowing for immediate alerts to be sent to the authorities in case of any abnormal health conditions. Additionally, the system monitors toxic gas levels, ensuring that the environment within the manhole remains safe. In the event of toxic gas detection, the system can activate neutralization mechanisms and send alerts to the concerned authorities for immediate action. This dual functionality ensures a safer working environment for maintenance workers, reduces the risk of accidents due to toxic gas exposure, and allows for quicker responses to emergencies. The implementation of this IoT-based monitoring system not only enhances worker safety but also provides a more efficient and proactive approach to managing underground maintenance operations, ultimately contributing to better urban infrastructure management. This project successfully designs and implements an IoT-based system to continuously monitor the health and safety of manhole maintenance workers while also ensuring the detection and neutralization of hazardous gases in the manhole environment. The dual-purpose system integrates advanced IoT technology, real-time data transmission, and automated alert mechanisms to address two critical challenges: worker health and environmental safety.

The health monitoring component of the system equips workers with wearable devices that measure vital signs such as heart rate, body temperature, oxygen saturation, and physical activity. These measurements are continuously relayed to a central monitoring station. If any abnormal health indicators, such as a drop in oxygen levels or an elevated heart rate, are detected, the system immediately triggers an alert to notify authorities, enabling them to take rapid action. This aspect ensures that even in confined spaces, where medical attention might be delayed, authorities are aware of potential health emergencies in real time.

The environmental monitoring component of the system continuously tracks the concentration of toxic gases such as methane, carbon monoxide, hydrogen sulfide, and others. Upon detection of unsafe levels, the system can initiate automated neutralization processes, including ventilation or chemical neutralization. Alerts are sent to the concerned authorities, highlighting the severity of the situation so that further remedial actions can be undertaken immediately. This real-time gas monitoring and neutralization system is crucial, given that manhole environments are notorious for toxic gas accumulation, which poses severe risks to human life.

The system architecture is designed with scalability and robustness in mind. By leveraging IoT infrastructure, the system is capable of covering multiple manholes across a city, providing authorities with a centralized dashboard for real-time monitoring and historical data analysis. This data can also be used to optimize maintenance schedules, prevent hazardous incidents before they occur, and enhance predictive maintenance techniques, ultimately contributing to more efficient urban infrastructure management. In terms of practical implications, the system has the potential to revolutionize manhole maintenance procedures, transforming them from high risk manual operations to data-driven, automated processes. The system can be scaled and adapted to various other industrial applications where workers operate in hazardous or confined environments, such as in sewers, mines, and industrial plants.

The IoT-based manhole monitoring and health tracking system represents a major step forward in occupational safety and infrastructure management. By enhancing the safety of maintenance workers and automating environmental hazard detection and response, this project delivers a critical solution to a long-standing challenge in urban and industrial operations. It offers a reliable and scalable way to mitigate risks, reduce operational hazards, and ensure a healthier, more secure environment for workers, all while optimizing resource management for city authorities.

6.2 Future Scope

The IoT-based manhole maintenance worker monitoring and toxic gas detection system holds immense potential for future developments and enhancements. As technology evolves, the system can be equipped with more advanced health monitoring devices, such as wearables that track not only vital signs but also additional health metrics like blood pressure, glucose levels, and stress indicators. These improvements could lead to early detection of health issues through AI-driven predictive analytics, enabling authorities to prevent emergencies before they occur. The integration of smart helmets and augmented reality displays could further empower workers by providing real-time data on their health and environment, making their tasks safer and more efficient.

Another exciting avenue for future growth lies in the system's integration with smart city infrastructures. As cities become increasingly connected, this system can be incorporated into a centralized platform that monitors multiple manholes across large urban areas. This would enable authorities to manage maintenance operations, detect hazards, and respond to emergencies with greater efficiency. By combining environmental monitoring with smart city technologies, cities can ensure not only the safety of workers but also optimize resources, thereby improving overall urban infrastructure management.

The environmental monitoring capabilities of the system can also be expanded to include additional sensors for other hazardous conditions, such as temperature, humidity, water levels, and radiation, making it applicable in a wider range of environments. The implementation of predictive analytics, driven by machine learning, could forecast gas buildup and dangerous conditions before they become life-threatening. Further advancements in neutralization technologies could lead to fully autonomous gas dispersion and hazard mitigation systems, reducing the need for human intervention in dangerous situations.

In terms of technological innovation, integrating the system with autonomous robots or drones for manhole inspections and hazard neutralization could transform how maintenance tasks are performed. These robots could be deployed to assess dangerous situations and perform routine checks without endangering human lives. Edge computing could also play a significant role in improving the system's real-time decision-making capabilities by processing data locally, reducing the time between hazard detection and response.

There is also significant scope for improving worker training and safety protocols through virtual reality and augmented reality simulations. These tools can provide immersive training environments where workers can practice responding to hazardous conditions in a controlled setting. AR can further be used in the field to provide workers with real-time guidance on safety procedures, equipment handling, or toxic gas neutralization, enhancing their ability to handle emergencies more effectively.

Scalability and cost optimization are also crucial considerations for the future. As IoT technology becomes more affordable, the system can be scaled up to cover a larger number of manholes and similar hazardous environments. This would make it more accessible, especially in developing countries where infrastructure may be lacking, yet worker safety is still a pressing concern. Advances in renewable energy-powered sensors and low-cost components could further reduce the system's operational costs, making widespread deployment more feasible.

The system has the potential to significantly influence policy and regulatory frameworks. As more data is collected on worker health and environmental conditions in confined spaces, it can inform policymakers and regulators, leading to improved safety standards and enforcement in industries that involve high-risk environments. The system can also serve as a benchmark solution to meet occupational health and safety compliance, ensuring that workers are better protected globally.

Through the integration of advanced IoT technologies, smart city systems, and AI-driven analytics, the system has the potential to drastically improve worker safety, enhance urban infrastructure management, and optimize hazardous environment monitoring. As new technologies emerge, this system could become a cornerstone in occupational safety, providing proactive solutions to protect workers and mitigate risks in a variety of dangerous settings.

REFERENCES

[1] W. Dargie, J. Wen, L. A. Panes-Ruiz, L. Riemenschneider, B. Ibarlucea, and G. Cuniberti, "Monitoring toxic gases using nanotechnology and wireless sensor networks," *IEEE Sensors Journal*, vol. 23, no. 11, pp. 12274–12283, 2023. DOI: 10.1109/JSEN.2023.3269723.

[2] A. I. Siam, M. A. El-Affendi, A. Abou Elazm, G. M. El-Banby, N. A. El-Bahnasawy, F. E. Abd El-Samie, and A. A. Abd El-Latif, "Portable and real-time iot-based healthcare monitoring system for daily medical applications," *IEEE Transactions on Computational Social Systems*, vol. 10, no. 4, pp. 1629–1641, 2022. DOI: 10.1109/TCSS.2022.3184519.

[3] S. Salehin *et al.*, "An iot based proposed system for monitoring manhole in context of bangladesh," in *2018 4th International Conference on Electrical Engineering and Information & Communication Technology (iCEEiCT)*, pp. 411–415, IEEE, 2018. DOI: 10.1109/CEEICT.2018.8628091.

[4] S. K. Andrews, A. V. Benevent Raj, M. Swedha, G. Preetha, R. S. Devi, and K. L. Narayanan, "Internet of things enabled real-time drainage service hole management system," in *2023 7th International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, pp. 1791–1797, IEEE, 2023. DOI: 10.1109/ICECA58529.2023.10395270.

[5] R. Dronavalli, K. Seelam, P. Maganti, J. Gowineni, and S. D. Challamalla, "Iot-based automatic manhole observant for sewage worker's safety," in *2022 International Conference on Automation, Computing and Renewable Systems (ICACRS)*, pp. 310–316, 2022. DOI: 10.1109/ICACRS55517.2022.10029252.

[6] V. S. A. K. Vidhya Sree A., Sudarmani R., "Development of manhole cover detection and continuous monitoring of hazardous gases using wsn and iot," pp. 202–206, IEEE, 2022. DOI:10.1109/ICCMC53470.2022.9754094.

[7] C. Martinez, M. Perez, and J. Torres, "Iot-based real-time monitoring for sewer manhole safety: A smart city application," *Smart Cities*, vol. 5, no. 1, pp. 111–128, 2023. DOI:
10.3390/smartcities5010001.

[8] A. Shajkofci, "Correction of human forehead temperature variations measured by non-contact infrared thermometer," *IEEE Sensors Journal*, vol. 22, no. 17, pp. 16750–16755, 2021. DOI: 10.1109/JSEN.2021.3102565.

[9] A. Elsaadany, A. Sedky, and N. Elkholy, "A triggering mechanism for end-to-end iot ehealth system with connected ambulance vehicles," in *2020 8th International Conference on Information, Intelligence, Systems & Applications (IISA)*, pp. 1–6, IEEE, 2020. DOI: 10.1109/IISA.2017.8316405.

[10] C. Nwibor, S. Haxha, M. M. Ali, M. Sakel, A. R. Haxha, K. Saunders, and S. Nabakooza, "Remote health monitoring system for the estimation of blood pressure, heart rate, and blood oxygen saturation level," *IEEE Sensors Journal*, vol. 23, no. 5, pp. 5401–5411, 2023. DOI: 10.1109/JSEN.2023.10017171.

[11] J. Zhang and H. Yang, "A privacy-preserving remote heart rate abnormality monitoring system," *IEEE Access*, vol. 11, pp. 97089–97098, 2023. DOI: 10.1109/ACCESS.2023.3312549.

[12] A. Vyas and S. Pal, "Power saving approach of a smart watch for monitoring the heart rate of a runner," *IEEE Transactions on Consumer Electronics*, vol. 69, no. 3, pp. 490–498, 2023. DOI: 10.1109/TCE.2023.3308629.

[13] L. Shu, Y. Chen, Z. Sun, F. Tong, and M. Mukherjee, "Detecting the dangerous area of toxic gases with wireless sensor networks," *IEEE Transactions on Emerging Topics in Computing*, vol. 8, no. 1, pp. 137–147, 2020. DOI: 10.1109/TETC.2020.2981296.

[14] E. Chavez, R. Vidal, A. Tejada, and J. Grados, "Image processing for the detection and´ monitoring of toxic gases in confined environments: An approach for application in shipping containers in callao-peru," in *2020 6th International Conference on Mechatronics and Robotics Engineering (ICMRE)*, pp. 160–164, 2020. DOI: 10.1109/ICMRE49073.2020.9065152.

[15] V. S. A, S. R, V. S, and A. K, "Development of manhole cover detection and continuous monitoring of hazardous gases using wsn and iot," in *2022 6th International Conference on Computing Methodologies and Communication (ICCMC)*, pp. 202–206, 2022. DOI: 10.1109/ICCMC53470.2022.9754094.

[16] J. Baron, S. Kim, L. Tan, and J. Choi, "A real-time iot-enabled sewer manhole monitoring system," *IEEE Internet of Things Journal*, vol. 2, no. 1, pp. 123–130, 2015. DOI:

10.1109/JIOT.2014.2367881.

[17] S. I. Nahid and M. M. Khan, "Toxic gas sensor and temperature monitoring in industries using internet of things (iot)," in *2021 24th International Conference on Computer and Information Technology (ICCIT)*, pp. 1–6, 2021. DOI: 10.1109/ICCIT54785.2021.9689802.

[18] S.-D. Han and Y.-S. Sohn, "Thin film gas sensor for detection of toxic gases from microbial," in *2011 16th International Solid-State Sensors, Actuators and Microsystems Conference*, pp. 178–181, 2011. DOI: 10.1109/TRANSDUCERS.2011.5969248.

[19] Q. Xie, G. Wang, Z. Peng, and Y. Lian, "Machine learning methods for real-time blood pressure measurement based on photoplethysmography," in *2018 IEEE 23rd International Conference on Digital Signal Processing (DSP)*, pp. 1–5, IEEE, 2018. DOI: 10.1109/DSP.2018.8347188.

[20] T. Lili and H. Wei, "Portable ecg monitoring system design," in *2019 3rd International Conference on Electronic Information Technology and Computer Engineering (EITCE)*, pp. 1370–1373, 2019. DOI: 10.1109/EITCE47263.2019.9095135.

[21] A. A. Patil and M. J. Godse, "Wireless ecg monitoring system," in *2019 International Conference on Advances in Computing, Communication and Control (ICAC3)*, pp. 1–5, IEEE, 2019. DOI: 10.1109/ICAC347590.2019.9036806.

[22] S. Bhowmick, P. K. Kundu, and D. D. Mandal, "Iot assisted real time ppg monitoring system for health care application," in *2021 IEEE Second International Conference on Control, Measurement and Instrumentation (CMI)*, pp. 122–127, IEEE, 2021. DOI: 10.1109/CMI50323.2021.9362852.

[23] M. Fatangare, H. Ohal, and V. Rathod, "Smart health monitoring system using pulse and temperature sensor (sensor-based health observer)," in *2023 7th International Conference On Computing, Communication, Control And Automation (ICCUBEA)*, pp. 1–5, IEEE, 2023. DOI: 10.1109/ICCUBEA58933.2023.10391984.

[24] M. Yang, X. He, L. Gao, and L. Zhao, "An innovative system of health monitoring using mobile phones," in *2012 IEEE 14th International Conference on e-Health Networking, Applications and Services (Healthcom)*, pp. 379–382, IEEE, 2012. DOI: 10.1109/HealthCom.2012.6379442.

[25] A. Kumar, R. Gupta, A. Kumar, A. Aditya, and A. Rai, “Simulation and development of sewage cleaning monitoring robot using iot,” in *2023 International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES)*, pp. 536–541, 2023. DOI: 10.1109/CISES58720.2023.10183525.

[26] J. T. Mazumder, R. K. Jha, H. W. Kim, and S. S. Kim, “Capacitive toxic gas sensors based on oxide composites: A review,” 2023. DOI: 10.1109/JSEN.2023.3289835.

[27] S. Srivastava, T. Sharma, and M. Deshwal, “Study of nanostructured metal oxide semiconductor based gas sensors for toxic gas detection,” in *2023 International Conference on Sustainable Emerging Innovations in Engineering and Technology (ICSEIET)*, pp. 657–662, 2023. DOI: 10.1109/ICSEIET58677.2023.10303596.

[28] M. N. Tahir, D. Falana, K. O. Daniel, D. Arevalo, J. Guanipatin, and U. Hassan, “A wearable multipurpose toxic gas-monitoring device for industrial applications,” in *2023 IEEE 19th International Conference on Body Sensor Networks (BSN)*, pp. 1–4, 2023. DOI: 10.1109/BSN58485.2023.10331298.

[29] M. Slimani, C. Mer-Calfati, J.-P. Poli, F. Badets, E. Friedman, V. Rat, T. Laroche, and S. Saada, “A multi-surface acoustic wave sensor platform for the detection and identification of toxic gases,” in *2023 IEEE SENSORS*, pp. 1–4, 2023. DOI: 10.1109/SENSORS56945.2023.10324973.

[30] C. R. Sugato Ghosh, Hiranmay Saha, “Portable sensor array system for intelligent recognizer of manhole gas,” *2012 Sixth International Conference on Sensing Technology (ICST)*, pp. 589–594, 2012. DOI: 10.1109/ICSensT.2012.6461748.

[31] D. A. N. M. R. B. H. S. Sugato Ghosh, Indranil Das, “Development of selective and sensitive gas sensors for manhole gas detection,” *2016 10th International Conference on Sensing Technology (ICST)*, pp. 1–5, 2016. DOI: 10.1109/ICSensT.2016.7796220.

[32] J. Zhu, Y. Fu, Y. Xing, Y. Zhang, and Q. Qiao, “Multi-component toxic gas monitoring system based on internet of things,” in *2019 IEEE International Conference on Mechatronics and Automation (ICMA)*, pp. 769–773, 2019. DOI: 10.1109/ICMA.2019.8816612.

[33] D. Kim, S. Kim, J. An, and S. Kim, “A portable colorimetric array reader for toxic gas detection,” in *2017 ISOCS/IEEE International Symposium on Olfaction and Electronic Nose (ISOEN)*, pp. 1–3, 2017. DOI: 10.1109/ISOEN.2017.7968928.

[34] P. C. V. Chaganti and K. S. Sukesh, "Smart iot solutions for subsurface gas monitoring and safety in critical infrastructures," in *2023 2nd International Conference on Automation, Computing and Renewable Systems (ICACRS)*, pp. 303–308, 2023. DOI: 10.1109/ICACRS58579.2023.10404390.

[35] K. Lee, S. Park, and J. Nam, "Low-power wireless manhole cover monitoring system for smart cities," *Sensors*, vol. 19, no. 3, p. 801, 2019. DOI: 10.3390/s19030801.

[36] H. Huang, M. Lin, and H. Chen, "Wireless monitoring system for urban sewage manholes using lora technology," *Journal of Environmental Management*, vol. 289, p. 112524, 2021. DOI: 10.1016/j.jenvman.2021.112524.

[37] I. Gupta, M. Rao, and R. Das, "Predictive maintenance of manhole covers using iot and machine learning," in *2022 International Conference on Emerging Technologies (ICET)*, pp. 98–103, IEEE, 2022. DOI: 10.1109/ICET54785.2022.00019.

[38] R. Singh, A. Kumar, and P. Sharma, "Iot based smart sewage manhole cover monitoring and management," in *2020 International Conference on IoT and Application (ICIOTA)*, pp. 63–67, IEEE, 2020. DOI: 10.1109/ICIOTA49874.2020.9116725.

[39] Z. Ahmed, S. Hamid, and Y. Ibrahim, "Iot-enabled smart manhole management system for real-time status, water level, and gas detection," in *2023 4th International Conference on Smart Cities and Urban Analytics (ICU)*, pp. 132–137, IEEE, 2023. DOI: 10.1109/ICU56443.2023.10123154.

[40] A. Holovatyy, V. Teslyuk, M. Lobur, S. Pobereyko, and Y. Sokolovsky, "Development of arduino-based embedded system for detection of toxic gases in air," in *2018 IEEE 13th International Scientific and Technical Conference on Computer Sciences and Information Technologies (CSIT)*, pp. 139–142, 2018. DOI: 10.1109/STC-CSIT.2018.8526672.

[41] P. Jayant, E. Vincent, Mohana, M. Moharir, and A. K. A R, "Smart health monitoring and anomaly detection using internet of things (iot) and artificial intelligence (ai)," in *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*, pp. 479–485, IEEE, 2024. DOI: 10.1109/ICoICI62503.2024.10696283.

[42] J.-m. Hu, J. Li, and G.-h. Li, "Automobile anti-theft system based on gsm and gps module," in *2012 Fifth International Conference on Intelligent Networks and Intelligent Systems*, pp. 199–201, IEEE, 2012. DOI: 10.1109/ICINIS.2012.86.

[1] [9] [26] [10] [8] [13] [2] [12] [19] [11] [14] [26] [19] [5] [25] [27] [29] [28] [34] [18] [33] [40] [32] [17] [15] [11] [38] [7] [39] [16] [37] [36] [35] [30] [31] [6] [20] [3] [21] [4] [22] [24] [42] [23] [41]

APPENDIX A

PROGRAM

```
#define BLYNK_TEMPLATE_ID ""

#define BLYNK_TEMPLATE_NAME ""

#define BLYNK_AUTH_TOKEN ""

#include <Wire.h>

#include "MAX30105.h"

#include "heartRate.h"

#include <SoftwareSerial.h>

#include <OneWire.h>

#include <DallasTemperature.h>

#include <BlynkSimpleEsp8266.h> // Include Blynk library for ESP8266 char auth[] =

BLYNK_AUTH_TOKEN;

char ssid[] = "Joes phone";                    // Replace with your Wi-Fi SSID

char pass[] = "1234567i";                 // Replace with your Wi-Fi password
```

```
#define ONE_WIRE_BUS D3

#define TEMP_THRESHOLD 40.0

#define ECG_PIN A0

#define THRESHOLD 550


#define HR_LOW 60

#define HR_HIGH 120

#define SPO2_THRESHOLD 95


SoftwareSerial SIM800(D1, D2);


MAX30105 particleSensor;

OneWire oneWire(ONE_WIRE_BUS); DallasTemperature

sensors(&oneWire);


unsigned long lastBeat = 0;

float beatsPerMinute = 0;

float spo2Value = 0;


void setup() {

    Serial.begin(9600);
```

```
  SIM800.begin(9600); SIM800.println("AT+CMGF=1");
  delay(1000);
  Wire.begin(D4, D5);

  if (!particleSensor.begin(Wire, I2C_SPEED_FAST)) { Serial.println("MAX30102 sensor not found!");
    while (1);
  }
  particleSensor.setup();
  particleSensor.setPulseAmplitudeRed(0x1F);
  particleSensor.setPulseAmplitudeGreen(0);

  sensors.begin();

  Blynk.begin(auth, ssid, pass); // Initialize Blynk connection
}

void loop() {
  Blynk.run(); // Run Blynk in the loop to maintain connection

  sensors.requestTemperatures();
  float temperature = sensors.getTempCByIndex(0);
```

```
Serial.println(temperature);

if (temperature > TEMP_THRESHOLD) {

    sendAlertSMS("Alert: Temperature Warning!");

}

Blynk.virtualWrite(V0, temperature); // Send

temperature to Blynk (V0)

int ecgValue = analogRead(ECG_PIN);

if (ecgValue > THRESHOLD) {

    unsigned long currentTime = millis();

    if (currentTime - lastBeat > 300) {

                                    unsigned long interval = currentTime - lastBeat;

        lastBeat = currentTime;

        beatsPerMinute = 60000.0 / interval;

    }

}



Serial.print(ecgValue);

Serial.print(" ");

Serial.println(beatsPerMinute);
```

```
  if (beatsPerMinute > 0 && (beatsPerMinute < HR_LOW || beatsPerMinute >

    sendAlertSMS("Alert: Heart Rate Warning!");

  }

  Blynk.virtualWrite(V1, beatsPerMinute); // Send heart rate to Blynk (


  long irValue = particleSensor.getIR();

  if (checkForBeat(irValue)) {

    long delta = millis() - lastBeat;

    lastBeat = millis();

    beatsPerMinute = 60 / (delta / 1000.0);

  }

  spo2Value = calculateSpO2(); Serial.println(spo2Value);



  if (spo2Value < SPO2_THRESHOLD) {

    sendAlertSMS("Alert: SpO2 Warning! Low Oxygen Level");

  }

  Blynk.virtualWrite(V2, spo2Value); // Send SpO2 to Blynk (V2)


  delay(10);

}
```

```
float calculateSpO2() {
    return 95.0; }


void sendAlertSMS(String message) { SIM800.println("AT+CMGS=\"+918921125269\"");
    delay(1000); SIM800.print(message);
    delay(1000);


    SIM800.write(26);
}
```